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 SURVEY

A Comprehensive Survey on Integrated Photonic Switch Fabrics: From Device to Network Topology and Controller Design

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ABSTRACT Emerging applications such as Artificial Intelligence (AI), the Internet of Things (IoT), and Big Data are significantly increasing data traffic within data center networks and high-performance computing (HPC) systems. This surge in data flow is pushing the existing network infrastructures to their limits. To address the demands of this data-driven era, there is a critical need for network technologies and architectures that offer higher capacity, improved performance, and reduced transmission delays. Central to these networks are switches, which play a crucial role in directing data traffic between various nodes and clusters with minimal latency. These switches must manage large bandwidths and support numerous ports efficiently. As electronic switches approach limitations in energy efficiency and bandwidth scaling, optical circuit switching has emerged as a promising alternative. Optical switches are being explored for their potential to provide higher bandwidth and lower energy consumption in data center and HPC networks. This survey paper reviews the most recent advancements in various types of optical switches, including those based on Micro-Electromechanical Systems (MEMS), Silicon Photonics, and III-V interferometric switches, as well as wavelength-selective and hybrid solutions. It covers aspects such as device design, switch fabric topology, tuning mechanisms (e.g., thermo-optic and electro-optic), and control plane implementations. The paper highlights the potential of these technologies while also discussing some challenges that remain in this field for further advancement of optical switches.

INDEX TERMS Optical switching, integrated photonics, network topology, reconfiguration, control plane, data center.

I. INTRODUCTION

Optically interconnected networks offer a potential alternative for electronic networks in short and long-haul communication to eliminate the need for lossy optoelectronic and electro-optic (O-E-O) conversions [1]. In addition, the steady growth of data traffic in data center networks (DCNs) and high-performance computing (HPC) pushes the boundaries of electronically switched networks beyond their limits.

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The emergence of data-intensive applications, such as Artificial Intelligence (AI), the Internet of Things (IoT), and big data, has significantly influenced the evolution of communication networks. These applications demand high-bandwidth, low-latency solutions that traditional electronic networks struggle to meet. For instance, AI workloads involve large-scale distributed computations requiring high-throughput interconnects, while IoT ecosystems generate real-time data necessitating ultra-responsive and scalable networks. Similarly, big data applications rely on the seamless transmission of massive datasets across globally distributed systems, emphasizing the need for high-capacity and efficient interconnect solutions.

According to the Cisco Global Cloud Index forecast [2], by the end of 2021, the annual global data center IP traffic was expected to reach 19.5 Zettabytes, marking an approximate quadrupling compared to the data volumes of 2016. As of 2024, global internet users have continued to grow, reaching approximately 5.35 billion [3], highlighting a sustained increase in data consumption and traffic. Cisco's projections for 2025 further anticipate a substantial expansion in internet users, particularly from emerging markets, leading to even higher global data traffic [4]. This growth is expected to intensify the demand for efficient and scalable network solutions.

The increased data traffic and computing demand in data center and HPC system networks outpace the bandwidth growth of electronically switched application-specific integrated circuits (ASICs) [5]. According to Moore's law, the bandwidth of electronic switches doubles every two years at the same cost to keep up with increasing network demands [6]. However, the further emergence of data-intensive applications, such as Artificial Intelligence (AI), Internet of Things (IoT), big data, and hardware-accelerated applications [7], [8] faces challenges posed by Moore's law of networking. This highlights the necessity for high-throughput and low-latency networks, capable of accommodating the growing demands of these applications sustainably.

Although electronic packet switching is currently a practical solution to transfer data, applications running across DCNs need restricted quality-of-service (QoS) demands that require higher inter-rack bandwidth capacity [9]. On the other hand, in DCNs where only electrical packet switching (EPS), increasing the inter-rack bandwidth capacity is limited by pin density in a ball grid array (BGA) [5] leading to a mismatch between QoS demands and the inter-rack networks capabilities [1], [10]. This mismatch in network capabilities and QoS demands has motivated researchers to find a solution to better accommodate the growing demands for higher bandwidth by introducing optical circuit switches (OCS) networks to be able to handle larger amounts of data traffic with both high throughput and low latency.

OCS has been investigated extensively as a future-proof solution to the limitations in bandwidth that arise with traditional electronic switches. Optical switches, boasting high bandwidth capabilities, are especially promising due to their ability to work seamlessly regardless of the specific format or data rate [1], [5], [11], [12]. Other than the elimination of E-O-E conversions, OCS reduces the electronic circuitry dedicated to modulation formatting at transceivers that can reduce the cost and processing delay, reducing the mismatch between QoS and inter-rack network capabilities [1], [13], [14]. These endeavors have resulted in significant progress and achievements over the last decade. An illustrative instance of this progress is exemplified by Apollo, the world's first large-scale production deployment of optical circuit switches (OCSEs) for data center networking [15].

In recent years, there has been a surge of interest in developing innovative optical switch technologies, with a focus on various materials and switch network architectures. These advancements encompass silicon-based switches [14], [16], III-V-based optical switches [17], [18], micro-electromechanical systems (MEMS) switches [19], [20], [21], liquid crystal on silicon (LOCOS) switches [22], [23], and even hybrid approaches that combine two or more of these methods [24], [25] which are controlled by diverse tuning mechanisms, each allowing for a dynamic alteration of the switching state [26], [27]. Despite the potential optical switching technologies offer, it is important to note that their overall performance can be constrained by challenges in their control plane [28], [29] (i.e., the electronic controller responsible for managing switch access and reconfiguring underlying devices within an optical switch to connect input ports to output ports). Therefore, ongoing research and development efforts are focused not only on refining the physical components but also on optimizing the control mechanisms to unlock the full potential of these advanced optical switch technologies [28], [29], [30].

This survey paper endeavors to encapsulate the recent advancements in optical switching technology, extending from the intricacies of device fabrication and tuning mechanisms to the broader perspectives of network topologies and control plane architecture. While several surveys published in recent years, such as [14], [31], [32], [33], that have explored various dimensions of optical switching technologies, their investigations predominantly concentrate on either a singular technology or methodology. Similarly, works like [34], [35], and [36] look into the applications of optical switches across diverse systems. However, these contributions, though valuable, often exhibit a narrowed focus either on specific technological facets or application-driven analyses, leading to a compartmentalized understanding of the field.

In contrast, our survey aims to bridge these gaps by offering a comprehensive synthesis that not only highlights individual technologies and methods but also interlinks them within the broader ecosystem of optical networking. By doing so, it seeks to provide a holistic view that encapsulates the multifaceted challenges and solutions from the device level to system-level implementations. Our goal is to furnish researchers with a unified narrative that maintains the cohesiveness of the subject matter, thereby facilitating a deeper understanding of both existing challenges and potential pathways for innovation in optical switching technologies.

In the following section, we provide a technical survey of available optical switching technologies and their principle of operation including Mach-Zehnder interferometers (MZI) [37], [38], [39], [40], micro-ring resonators (MRR) [41], [42], [43], grating structures [44], [45], [46], [47], [48], and hybrid solutions [24], [25], [49], [50], [51], in three main categories of silicon-on-insulator (SOI), III-V materials, and III-V/Si hybrid designs. We also discuss the potential and impact of emerging technologies such as silicon nitride

(SiN), MEMS, and Liquid Crystal on Silicon (LCOS) in Section II. In Section III, we explore the different tuning mechanisms required for elastic switching elements, along with recent advancements in this area. Section III also looks further into the discussion of different optical switch tuning mechanisms including thermo-optic tuning, electro-optic tuning, phase-change materials, and all-optical switches. In Section IV, we explore various network topologies and the challenges involved in designing them. Following this, in Section V, we investigate controller design and different control schemes. By the end of each section, we also provide an overview of the key challenges and research opportunities for the respective area, enabling researchers to understand the field comprehensively.

II. OPTICAL SWITCHING DEVICES

Optical switching devices are the cornerstone of optical communication networks to realize high-speed data communication. These devices can steer the direction of light propagation. A network of interconnected switching devices, which we will discuss these networks in the topology section, that is orchestrated by a controller, discussed in the controller section, creates a switch fabric that can be reconfigured to direct optical signals from an input port to output, allowing for flexible and efficient management of optical traffic.

Optical switches are available in various forms, including MEMS, LCOS, and Silicon Photonics (SiPh) switches, among others. While this paper primarily focuses on integrated solutions, it also touches upon discrete solutions. Each type of switch offers unique advantages and limitations, necessitating careful selection based on the specific needs of the application. In the subsequent sections, we delve deeper into the characteristics and functionalities of different optical switching devices.

A. SILICON-PHOTONIC-BASED SWITCHING DEVICES

SiPh devices are extensively explored due to their adaptability to complementary metal oxide semiconductor (CMOS) technology, making them an attractive option for optical switching applications to be integrated with electronic circuitry. The SOI platform has matured over the past few decades and can be manufactured using CMOS technology, enabling cost-effective mass production of SiPh switches. Additionally, silicon has strong thermo-optic and free-carrier plasma dispersion coefficients, facilitating reconfiguration solutions to enable optical switching. Reconfiguration is usually done by manipulating the refractive index of Silicon waveguides using an actuator that is discussed in tuning mechanisms. This allows the switch to direct an incoming optical signal to any of several output waveguides [12], [14].

In this section, we will discuss the operation of building blocks that are used for switching in the SiPh platform, and in each section, we will review some of the most recent switch fabrics that are realized using these switching devices.

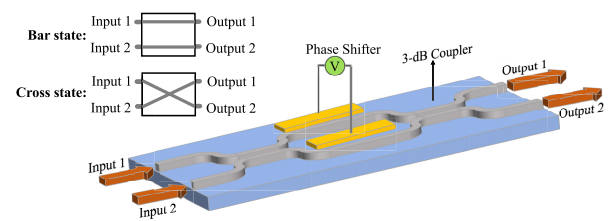


FIGURE 1. Schematic of a SiPh 2×2 MZI switching element. The tuning mechanism can switch the input light to either of the two outputs.

1) MACH-ZEHNDER INTERFEROMETERS (MZIs)

MZI is one of the most well-known switching devices that have been explored on different platforms including SiPh. Fig. 1 demonstrates the schematic of a 2×2 MZI switching element [52]. An MZI switching element consists of two 3-dB couplers and two waveguide arms connecting them. It operates by splitting the input signal into two paths and then recombining them at the output coupler. The relative phase between paths determines the output port. By controlling the phase difference between the two beams on the arms, it is possible to control whether the beams interfere constructively or destructively at the output 3-dB coupler by inducing a phase shift on MZI arms as the phase difference changes from 0 to π , resulting in switching light between the two outputs [12].

As an example in Fig. 1, when the light enters the device from Input 1, if no phase difference is introduced by the phase shifter located on the upper arm, then the light signal is transmitted to Output 2 (Cross-state). In a case when a phase shift is introduced by a tuning mechanism (discussed in section II), the switch is now in Bar-state and the signal is present at Output 1 (see Fig. 1).

The phase shifter can be designed variously to introduce the phase difference between the two arms of the MZI switch. These techniques are known as tuning mechanisms, such as electro thermo-Optic (TO), electro-optic (E-O), and electroabsorption [53], which is covered in the following section: tuning mechanisms.

Losses in an MZI can be caused by two main factors: phase error, which is caused by the imperfect phase between the two arms, and coupling imbalance between the two arms. These two factors are the main source of loss and crosstalk in MZIs [12], [54]. As a result, many research efforts have focused on designing high bandwidth 3-dB couplers for MZI switching elements and compensation of uneven phase shift [11], [54].

Examples of some of these efforts for the realization of 2×2 power splitters, including adiabatic directional couplers (DC), bent DCs, and phase-modulated DCs [39], [55], [56], [57], [58], [59], [60]. The work in [39] focuses on the reduction of random phase errors by the deployment of varied-width multimode interference (MMI) couplers in their MZI design. Inverse design approaches based on machine learning are deployed in some work such as [61], [62], [63]

to inverse design the structures that can have wider band response and lower phase error. The work in [64] used tapered Euler S-bends with mode filters to reduce the random phase imbalance in MZI switching elements. In a more recent work in [65] authors developed a pair of differential E-O phase shifters that enable arbitrary phase tuning while maintaining balanced free-carrier-absorption (FCA) loss across the MZI arms.

Crosstalk in MZI structures can be reduced by using one port of the MZI in network topologies which can be seen as 1×2 and 2×1 switching elements to limit the interference of the two optical signals in the two arms [66]. Another solution involves using variable couplers [67] which can be achieved with another two interferometers as discussed in [68] and [69] that only needs $\pi/2$ phase shift, unlike conventional MZI structures, this type of devices use two electrical signals to control the behavior of light as it travels through the device. By applying a push-pull signal to the two arms of the MZI, the phase difference between the arms can be controlled, which affects the interference at the output. This can help balance the insertion loss between the Bar and Cross state, as it helps reduce the amount of optical attenuation that occurs in one arm of the MZI [11]. It is worth noting that push-pull MZI switches have some advantages over conventional MZI switches, such as lower drive voltage and faster response time. However, they also have some limitations, such as more complex fabrication requirements and narrower bandwidth [11].

MZI switching elements are usually configured by either TO or EO tuning as shown in [40] and [54]. For example, in [54], a TO MZI switching element with adiabatic bends and integrated n-doped Silicon heaters is introduced. It has a switching time of $2.4 \mu\text{s}$, a small footprint of approximately $10 \mu\text{m}$ in length, and a power consumption of $12.7 \text{ mW}/\pi$. However, the operating voltage of this switch is reported as 12 V , which is challenging in a standard CMOS foundry process.

As an example of EO tuning, the work in [40] introduced a 2×2 EO-tuned MZI switching element with a switching power of 0.6 mW and a drive voltage of 0.83 V . The reported switching time of this element is 6 ns , and the crosstalk is less than -17 dB over an optical bandwidth of 60 nm .

It is also possible to design switching elements controlled by both TO and EO tuning mechanisms using a double-layer network (DLN) architecture. This incorporates microheaters and p-i-n diodes into the MZI waveguides [70]. By integrating these components, the DLN architecture enables simultaneous control of the refractive index and absorption coefficient of the waveguides, facilitating dynamic control of optical switches for various applications. In [70], a 4×4 optical switch based on the DLN structure was introduced. It is composed of twelve 2×2 MZI switching elements equipped with both TO and EO tuning mechanisms. The power consumption and switching delay for the TO and EO modes are reported as $34 \text{ mW}/\pi$, $72 \mu\text{s}$, and $7 \text{ mW}/\pi$, 5.7 ns , respectively.

A comparison between different proposed MZI-based switch cells is presented in Table 1, showcasing key parameters such as switching time, power penalty, bandwidth, cross-talk, extinction ratio, loss, and footprint. The table highlights the variations in design approaches, MMI, T-O modulation, and WDM, along with their performance metrics across C-band applications. This comprehensive comparison provides insight into the trade-offs involved in MZI-based switch designs, catering to diverse use cases.

2) MICRORING RESONATORS (MRRS)

In an MRR switching element based on an add-drop structure (see Fig. 2), the input light can be coupled into the ring for certain wavelengths (λ_r) and traverses the ring multiple times. The coupled light is built by altering the refractive index of the MRR, the resonant wavelengths can be changed, allowing the input light to be switched to either through or drop port based on the alignment with the resonance of the MRR [75].

One of the most important differences between MRRs and MZIs is that MZI switches operate based on interference but MRR switches operate based on resonance. When the optical signal's wavelength matches the resonant wavelength of the ring, the signal is coupled from the bus waveguide into the ring, effectively "switching" the signal's path [14]. By adjusting the phase shift and resonant wavelength in the MRR, the incoming light can be switched from one of the input waveguides to either of the output waveguides, depending on the state of the switch [76].

The compactness of MRRs makes them an attractive option for integration into SiPh platforms, for use in optical switch networks [14], [76]. On the other hand, one of the main challenges of using MRRs is that they are sensitive to temperature changes, which can cause changes in the MRR's resonant wavelength. Furthermore, MRR performance can be significantly impacted by small fabrication errors [11], [42], [76]. To reduce this error researchers have discussed different designs to improve the performance of MRRs under fabrication variations [38], [77].

The most simple form of an MRR-based switch configuration can be created by placing the ring in the proximity of two parallel waveguides (i.e., input and drop waveguides), as shown in Fig. 2(a). Tuning can be achieved through E-O or T-O mechanisms (see Section III).

Fig. 2(b) demonstrates an embedded PIN junction integrated with the ring structure, as implemented in [78], where cascaded microring resonators were used to realize high-speed electro-optic switching through carrier-injection modulation. The device achieved 100 Mbps operation using forward-biased PIN diodes to tune the resonant wavelengths, demonstrating efficient and compact switching on a CMOS-compatible platform. In this configuration, the wavelength will be blocked when the working wavelength aligns with the MRR's resonant wavelength. In an add-drop MRR configuration, the two waveguides can be arranged in a crossing

TABLE 1. MZI-based switches comparison.

Ref.	Radix	Switching time	Power Penalty	Bandwidth	Cross-talk	Extinction Ratio	Loss	Footprint	Description
[71]	2×2	$0.7 \mu s$	1.85 mW	C-band	-28 dB	28.6 dB	0.88 dB	N/A	MMI and T-O
[72]	2×2	$0.4 \mu s$	23.2 mW	N/A	C-band	38 dB	0.9 dB	$60 \times 30 \mu m^2$	T-O doped MZI+heat sink
[73]	2×2	$530 \mu s$	0.48 mW	N/A	N/A	21 dB	1.5 dB	$420 \times 180 \mu m^2$	T-O + trench design
[74]	2×2	$21 \mu s$	117 mW	C-band	-18.8 dB	13.6 dB	1.4–10.8 dB	$740 \times 700 \mu m^2$	T-O + WDM and MDM
[67]	2×2	$10 \mu s$	N/A	C-band	-50 dB	50.4 dB	0.5 dB	N/A	T-O + variable splitter

(see Fig. 3(b)) or parallel (see Fig. 3(a)) configuration and are coupled to the ring. When the input signal is aligned with the resonance, it is routed to the drop port. Otherwise, it goes through the port by adjusting the wavelength alignment with the resonance of the MRR. An input optical signal can be directed to either the through or drop port, also known as bar and cross state [11], [14]. In [29] authors exploit the binary bar/cross behavior of each MRRs to minimize trimming power: after measuring fabrication-induced resonance shifts, PARS pre-assigns every ring a default drop- or through-state that yields the lowest steady-state consumption, then routes signals accordingly. The underlying controller algorithm and its power-saving impact are detailed in Section V.

To widen the resonance peak of an MRR, higher-order MRRs are usually used which makes capturing the high-speed data easier [14]. In these designs, more than one ring is used for creating a flat resonance peak as discussed in [79]. Fig. 4 shows an example of MRR based optical switch network in an Omega topology as discussed in [41] where dual add-drop MRRs are controlled independently for routing signal. When both MRRs were aligned in resonance they could achieve a 165 GHz operation which is higher than single MRR switch. As another example of second-order mode MRRs we can mention [80], where an optical switch array featuring 12 second-order MRR elements with a 50 GHz bandwidth, based on butterfly topology and controlled by the thermo-optic effect, is introduced.

To mention some of the research in this area, we can mention the work in [43] where the MRR switch utilizes the T-O effect of SiPh waveguides to heat the ring through heaters sitting on top of the ring (refer to Fig. 4). This paper reported that a small wavelength shift of approximately 3 nm and a high extinction ratio (ER) of greater than 20 dB can be achieved within a 30 nm wavelength range. In [80], a 4non-blocking switch with butterfly structure was introduced based on dual MRR. We will discuss topologies in a dedicated section, however, among different topologies butterfly is among the non-blocking ones that use less switch elements. In [81] presents the development of an ultra-low-loss multi-layer 8×8 microring optical switch leveraging a three-dimensional (3D) microring-based optical switch element on a multi-layer Si₃N₄-on-SOI platform. The

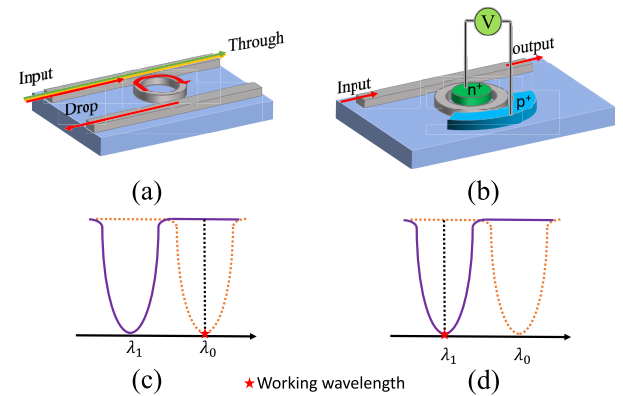


FIGURE 2. (a) Demonstration of an MRR with one waveguide. (b) A tunable MRR switch with a PIN junction for E-O tuning. (c) Transmission spectra of the MRR switch when the switch status changes from OFF to ON state. In this demonstration, λ_0 and λ_1 are the resonant wavelength without and after modulation, respectively. The orange dotted line represents the operation of the switch when the logic signal is '1' and the purple line is the spectra when the logic signal is '0'. (d) Transmission spectra of the switch when it switches from the ON to OFF state [29]. Copyright ©2023, IEEE.

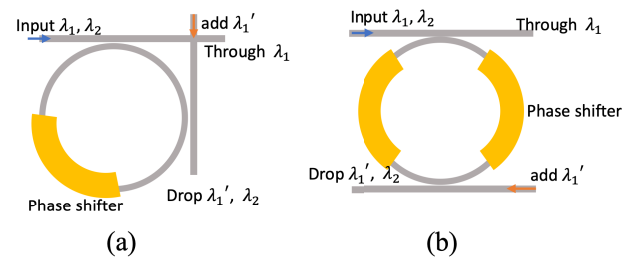


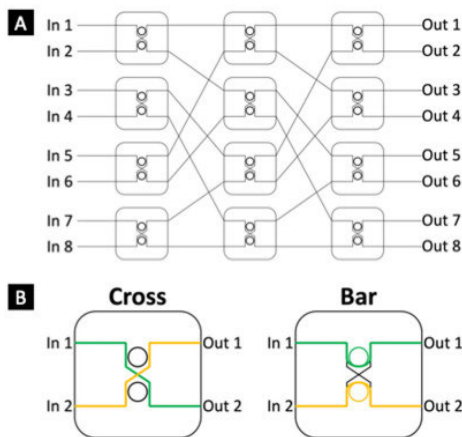
FIGURE 3. (a) An MRR with crossing outputs. (b) MRR with parallel outputs and phase shifters are placed on the ring. Figure redrawn from [11].

switch features Si/Si₃N₄ waveguide overpasses and Si₃N₄ dual-coupled microring resonators, achieving low insertion loss and calibration-free operation. Its design minimizes the complexity of switch control by ensuring that only one switching element needs to be active in each routing path.

In Tabel 2 we compare different proposed MRR-based switches proposed in the literature.

TABLE 2. MRR-based switches comparison.

Ref.	Radix	Switching time(Phase shifter)	Power Penalty	Bandwidth	Crosstalk	Extinction Ratio	Loss	Footprint	Description
[41]	8×8	0.4 - 1.2 μ s (T-O)	18.7 mW	55 GHz	-16.75 dB	14.7 - 18.8 dB	4.4 - 9.6 dB	4 mm ²	Omega
[81]	8×8	N/A	N/A	~69 GHz	< -15 dB	N/A	0.52-2.66 dB (on-chip)	3.32× 3.62 mm ²	crossbar+multi-layer
[82]	4×4	19.2 μ s (T-O)	47.46 mW (on average)	50 GHz	<-28 dB	N/A	< 2.5 dB	176 × 632 μ m ²	Half-Crossbar
[80]	4×4	- (T-O)	172.6 dB	50 GHz	-21 - -6.6 dB	-	< 5 dB	<2.5 mm ²	Butterfly
[83]	4×4	- (T-O)	-	46 GHz	-13.6 dB	-	<= 6.9 dB	3.47 × 2.43 mm ²	Benes
[84]	4×4	- (T-O)	-	24 GHz	-	-	0.8 - 4 dB	-	Switch-and-Select
[85]	2×2	1 ns - (E-O)	37 mW	100 GHz	-	50 dB	3.1 - 5.9 dB	-	-
[86]	4×4	25 μ s - (T-O)	- (23 fJ/bit)	-	-	50 dB	-	-	-
[87]	16×16	- (T-O)	< 10 dB	70 - 100 GHz	< -39 dB	20 - 22 dB	0.1 - 0.2 dB	-	Clos


FIGURE 4. Demonstration of the Omega switch network as described in [41]. Figure reprinted from [41] Copyright ©2020, IEEE.

3) GRATING STRUCTURES

Grating-assisted structures are essential components used for manipulating light properties to perform various functions such as wavelength filtering, modulation, and routing. The fundamental unit of SiPh grating is a periodic modulation of the refractive index achieved either by varying the refractive index or the physical dimensions of the waveguide, leading to a series of reflections. These reflections positively overlap within a limited range of wavelengths around a specific wavelength known as the Bragg wavelength [88]. By applying a control signal, the refractive index of the waveguide material changes, which alters the wavelength of light reflected by the Bragg grating [45].

Integration of Bragg gratings was first demonstrated in [89]. The early integrated Bragg grating structures were

based on wide rib waveguides [89], [90], [91]. However, more recently researchers are more focused on the use of strip waveguides. Small side wall corrugation in the submicron strip waveguides can create a significant grating coupling coefficient resulting in large bandwidth devices up to hundreds of tens of nanometers [92]. The grating structure can be designed for WDM applications also to have narrower bandwidths such as what has been discussed in [46] where they achieved 3 dB bandwidth smaller than 50 GHz by deploying 1200 nm wide waveguides and third order gratings.

Waveguide gratings are commonly used in reflection mode as 2-port devices, however, this requires the use of an optical circulator, which increases cost and complexity. Grating-assisted asymmetric waveguide couplers have a very weak reflection at the operating wavelength, allowing them to function as 4-port devices without the need for optical isolators or circulators, making them a simpler and more cost-effective solution [45]. Grating-assisted asymmetric waveguide couplers which are also known as Contra-directional couplers (contra-DCs or CDCs) have been the subject of many research studies in recent years due to their ability to function as wavelength-selective filters and switch structures [45], [48], [93]. CDCs are designed to couple light in opposite directions. They consist of two parallel waveguides, similar to directional couplers, with a grating pattern etched on the side of the waveguides that is designed to couple light into the other waveguide. However, the structure of a CDC differs from a normal DC in two important ways. The waveguides in contra-directional couplers have different widths, which results in different propagation constants in each waveguide and reduces the normal coupling in the same direction. Additionally, a repeating dielectric perturbation is used, which causes a specific wavelength of light to be transferred into the other waveguide in the opposite direction, similar

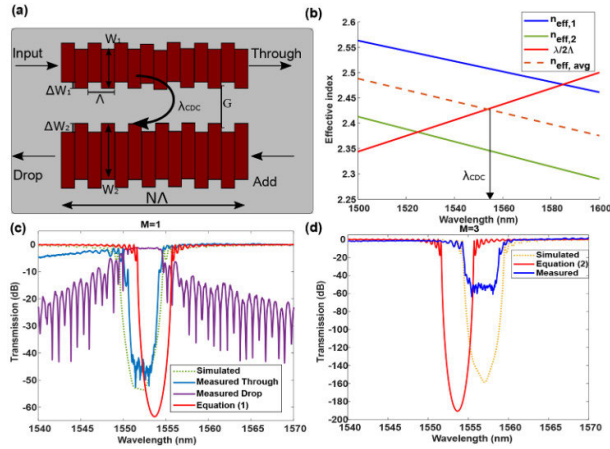


FIGURE 5. (a) Demonstration of a CDC structure. The waveguides have different widths to minimize the directional coupling effect between the two waveguides. Since this structure has four ports so it can eliminate the need for an optical circulator. (b) Demonstration of phase match condition. The drop port wavelength is determined by the intersection of the average effective index of the two waveguides and the $\lambda/2\Lambda$. (c) Drop- and through-port responses based on different methods of calculation and the equation described in [44] (d) Response from a three-stage cascaded CDC filter (for more detail see [44]). Figure reprinted from A. E. Afifi, M. Hammood, N. A. F. Jaeger, S. Shekhar, L. Chrostowski, and J. F. Young, 'Contra-directional pump reject filters integrated with a micro-ring resonator photon-pair source in silicon,' *Optics Express*, vol. 29, no. 16, pp. 25173-25188, 2021. Licensed under a Creative Commons Attribution (CC BY) License.

to the way Bragg gratings work [45]. Figure 5 demonstrates a CDC structure and the response of each port. When the wavelength λ_D satisfies the Bragg condition ($\lambda_{CDC} = (n_1 + n_2)/\Lambda$, where the n_1 and n_2 are the effective indices of the waveguides and Λ is the grating period of the structure) the light signal is redirected to the drop port of the CDC structure.

Conventional strip waveguide-based CDCs typically don't provide bandwidth greater than 10 nm [94]. To effectively separate the C-band from other wavelength bands in WDM applications, it is crucial to have bandwidths that exceed 30 nm. Consequently, subwavelength gratings (SWGs) can be used to increase the bandwidth of CDCs [94], [95], [96], [97]. SWG are periodic structures where the period of the grating is smaller than the wavelength of the light, so it doesn't cause Bragg diffraction [98]. SWGs can be designed to have less mode confinement which can contribute to a higher CD-coupling coefficient and increase the bandwidth of SWG devices [94]. Additionally, the proposed structures based on strip waveguides have been demonstrated to reach relatively high 3 dB bandwidths however they suffer from low sidelobe suppression ratios (SLSRs) in their passband (near 3 dB) [95]. Apodization is another method that has been deployed for better SLSRs. Apodization in grating structures refers to the gradual variation of the grating's depth across its width. This is often used to reduce the sidelobes in the diffraction pattern produced by the grating, which can improve the resolution and signal-to-noise ratio of the system [48], [96], [97].

In [99] an SWG CDC with the bandwidth of 4 THz was achieved over the length of 150 μm however, the

proposed structure had a minimum feature size of 40 nm which adds complexity to the fabrication of this device. Yun et al. in [94] proposed an SWG CDC with an SLSR of 19 dB and a bandwidth of 32.6 nm with a larger minimum feature size of 60 nm. Similar to MRR and MZI structures CDC drop port response wavelength can be controlled with T-O or E-O effect. A silicon tuneable add-drop filter with a segmented microheater based on CDCs is discussed in [47] where by applying appropriate temperature offsets and slopes, the filter's center wavelength can be continuously tuned over 8nm, and its drop-port 3dB bandwidth can be adjusted between 14.0 and 22.4nm. Similarly, in [100], an ultracompact, bandwidth-tunable filter was demonstrated using a silicon-on-insulator wafer, which utilizes a combination of cascaded grating-assisted contra-directional couplers (GACDCs) and a subwavelength grating (SWG) structure. By applying heat to one of the GACDCs, a tunable bandwidth of 6 nm was achieved. However, cascading CDC causes the non-overlapping spectral regions of the response to create a residue that can't be integrated back into the main response [101]. In [47] authors proposed a thermally chirped structure to increase the bandwidth of a CDC filter without cascading. In the proposed solution eight different heaters were leveraged to create different offsets and slopes to create by just using the T-O coefficient. This method however can face limitations in the suppression of sidelobes when the slope of the heating power increases.

Due to the non-interferometric nature of device grating structures, creating wide band optical switches with them is challenging. In tuning mechanisms (section III) we will discuss methods proposed in literature to deal with this problem.

4) HYBRID STRUCTURES

We discussed the most fundamental building blocks that are used for the design of switching elements in the previous sections including MRR, MZI, and grating structures here we mention some of the efforts that have been made to propose novel switching elements based on a combination of these structures.

Devices discussed in this section are innovative methods that combine devices we discussed earlier to achieve switches tailored for special use cases. As an example, [51] reported a 2×2 silicon E-O switch based on the double-ring assisted MZI (DR-MZI) where they experimentally demonstrated an active switching element with a bandwidth of 60 GHz and crosstalk of -20 dB and switching power of 3 mw. The device created by MRR and an MZI uses one MRR on the upper arm of the MZI at a resonance wavelength of λ_1 and another MRR on the bottom arm that has its resonance wavelength blue shifted to λ_2 after tuning. By utilizing the benefits of both MRRs and MZIs, such as resonance enhancement and coherent interference, a small change in the refractive index of one ring can lead to a π phase difference between the two arms allowing for a

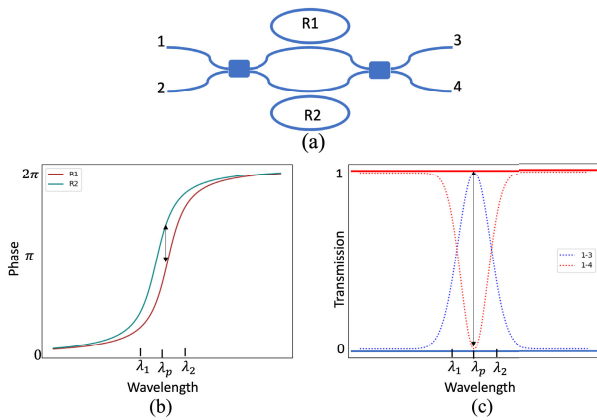


FIGURE 6. (a) A schematic representation of the 2×2 DR-MZI-based EO switch. (b) Phase shifts for the two MRRs, with one ring (R1) blue-shifted after tuning. (c) Transmission spectra for paths 1–4 and 1–3, where solid lines indicate spectra before tuning and dotted lines represent spectra after tuning. Redrawn figure from [51].

more power-efficient switching mechanism either individual MRRs or MZIs. Similarly, in more recent work in [69] a $2 \times 2 \times 2\lambda$ wavelength selective switch is proposed based on a combination of MZI and MRRs. The design includes multiple pairs of over-coupled microring resonators as efficient and narrowband phase shifters, which were driven in a push-pull scheme. Where they could achieve cross-talk suppression and switching extinction ratio of over 21 dB. The proposed structure in [51] is demonstrated in Fig.6.

Ikeda et al. [25] demonstrated a 16 channel 2×2 SOI wavelength wavelength-selective switch based on a combination of T-O MZI and CDCs. In the proposed architecture sixteen 2×2 single channel wavelength selective switches (WSS) are cascaded. The 16-channel averages for insertion loss, on-chip loss, 3-dB bandwidth, bar-port extinction, and cross-port extinction were 9.2 dB, 5.4 dB, 4.2 nm, 15.0 dB, and 23.0 dB respectively and the reported average trimming power and switching power were 3.8 mW and 15.6 mW respectively. The idea of this switch was discussed in [50] for one and two-channel wavelength selective switch structures in wavelength channels of 1532 and 1553 nm. Fig. 7 from [50] demonstrates the fabricated 2×2 single channel wavelength selective switch and the different port connection spectra of the switching element. In this design, the CDCs function as a filter that can manipulate the signal and the MZI determines the output.

The idea of hybrid architectures based on CDCs and MRRs has been discussed in [49] and [102]. The combination of CDC and rings can eliminate the ring FSR. The main drawback of this method is the large footprint of the devices compared to the MRRs. In [103] authors replaced one of the CDCs from the race track resonator discussed in [49] with an MZI to add bandwidth tuneability to their filter design. In another work [24], a 16×16 switch based on a dual ring assisted MZI is introduced which is based on an MZI structure coupled with two micro rings on each

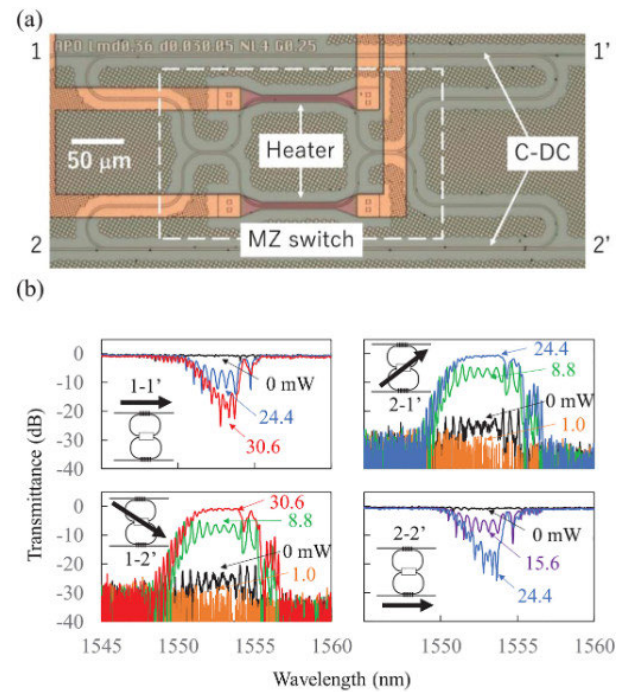


FIGURE 7. Structure of a CDC MZI structure as discussed in [50]. Reprinted from [50], Copyright ©2020, IEEE.

arm. Both TiN and PIN diodes are integrated into the switch for ring resonance alignment. The proposed switch uses Benes architecture and could achieve a 0.3 nm bandwidth. By adding the MRRs to the MZI arms it's possible to change the state of the switch by inducing a small resonant shift in MRRs, unlike normal MRRs where small refractive index modulation is not enough to cause shift and usually has larger insertion loss (IL) or lower ER. however, this method can achieve better ER at the cost of a larger footprint and more power consumption.

Combination devices can lead to new architectures that can have the benefits of both devices, however, at the cost of a larger footprint and more complex control. Recent advancements in hybrid structures include the integration of Co-Packaged Optics (CPO) with traditional photonic platforms. By embedding photonic and electronic components into a unified package, CPO-enabled hybrid architectures optimize the synergy between devices like MZIs and MRRs, offering seamless scaling for high-bandwidth applications. For instance, a CPO-based hybrid switch demonstrated in [104] integrates SiPh and MZI arrays, achieving over 60 GHz bandwidth and 25 dB crosstalk suppression. Furthermore, heterogeneous integration techniques are enhancing hybrid designs by combining III-V materials and silicon, which extends operational wavelength ranges and improves energy efficiency. These developments mark significant progress toward overcoming bottlenecks in traditional photonic architectures. Recent advancements in CPO and hybrid photonic integration have demonstrated the potential to achieve ultra-compact and efficient designs

for high-bandwidth applications. An illustrative example is the development of a 4×112 Gb/s polarization-insensitive SiPh WDM receiver hybridly integrated with a CMOS transimpedance amplifier (TIA) [105].

In Table 3 we have provided a comparison of different hybrid structures over the past few years.

B. III-V PHOTONIC

Over the past few decades, III-V technology has experienced rapid development. One of the major drawbacks of silicon devices is their inability to integrate with active devices due to silicon's indirect bandgap [107]. This limitation has been addressed by the integration of InP-based technology, which has enabled the development of highly complex on-chip systems featuring high-performance active components such as optical amplifiers and lasers [107], [108]. However, III-V technology lack scalability compared to SiPh.

Optical switches that we are going to review in this section are based on semiconductor Optical Amplifiers (SOA) that amplify an optical signal directly, without the need to convert it to an electrical signal first. It works by using a semiconductor as the gain medium, where the incoming light signal stimulates the emission of additional photons coherent with the input [109]. Integrated switch fabrics utilizing SOA-gate technology have primarily been designed in two configurations: broadcast-and-select and wavelength-selective. In SOA-based designs, each signal path is gated with a single SOA, which can typically perform switching for a single polarization. However, it is also possible to design polarization-insensitive SOAs. Due to the inherent gain of SOAs, these switches exhibit high crosstalk suppression and an impressive ON/OFF extinction ratio, which helps overcome losses [110], [111]. In the following section, we will discuss switch fabrics based on III-V technology, focusing on three main sub-categories: SOA-AWG, SOA-MZI, and SOA-MRRs.

1) SOA-AWG SWITCHES

One way to achieve WSS for devices using SOA gates is by incorporating arrayed waveguide gratings (AWGs) on the same chip. The AWGs act as built-in de-multiplexers, with each output port connected to an SOA gate that can be used to select and route specific wavelengths. In this configuration, the SOA elements are predominantly used for gating and the AWGR performs wavelength routing and the cooperation of these two elements works as an optical switching element [35], [112]. This method of switching provides advantages such as low switching time since the SOAs are relatively fast in switching and due to the inherent gain of SOAs switches cooperating SOA have higher extinction ratio and lower crosstalk. Further, SOAs possess a wide range of gain spectrum, unlike erbium-doped fiber amplifiers (EDFAs) which have a narrower gain spectrum, making them capable of signal amplification in many different wavelengths. When an SOA is subjected to high

input power, its gain becomes saturated. Gain saturation occurs because the number of available carriers in the active region of the SOA becomes depleted as they are used to amplify the input signal. This saturation effect helps to equalize the gain across multiple wavelengths [35], [113], [114]. The design of AWG switches is subjective to the number of waveguides available in AWG. The higher the number of waveguides, the more wavelengths can be routed by the optical switch. However, increasing the number of optical waveguides in AWG comes at the cost of a larger footprint of AWG and a more complex fabrication process. The waveguide spacing in AWG design is strongly correlated with the channel spacing of the AWGs. Tighter waveguide spacing results in a higher resolution, allowing for more precise wavelength routing and potentially supporting a larger number of channels within a given wavelength range. However, smaller waveguide spacing can also increase crosstalk between adjacent channels and require more precise fabrication techniques. The FSR range of AWGs is another aspect that needs to be considered when designing them. A larger FSR enables the AWG to support a broader wavelength range, allowing for more channels and improved system capacity. However, increasing the FSR requires a larger AWG chip size, which can impact the device's size, cost, and insertion loss. The insertion loss of an SOA-AWG switch depends on various factors, including the number of waveguides, waveguide spacing, and FSR. Generally, as the number of waveguides and the FSR increase, the insertion loss may also increase due to the larger device size and more complex fabrication process [35], [113], [114], [115].

A monolithic implementation of an InGaAsP/InP 4×4 wavelength selective cross-connect has been reported in [116] where the four inputs are distributed to two levels of gates that utilize semiconductor optical amplifiers, which are used for both wide-range input selection and selection based on specific wavelengths. A similar concept was deployed to realize a monolithically integrated InGaAsP/InP active-passive 8×8 with a switching time of 7 ns, which has been demonstrated in [117]. This method has also been deployed for the realization of Data center networks optical switching such as [118], [119], [120].

The essence of Sirius [119] is the utilization of a combination of adjustable lasers and straightforward gratings that direct light based on its wavelength. Sirius' switching technology, routing and scheduling, congestion control, and time-synchronization mechanisms are tightly integrated to create a flat and scalable network with high bandwidth and low latency. A small-scale prototype utilizing a customized laser chip that can tune in under 912 ps has been developed, demonstrating 3.84 ns end-to-end reconfiguration over 50 Gbps channels.

In [118] and [121] a wireless data center network-based architecture using an SOA-based wavelength switch is introduced. The network consists of N clusters, each connected by inter-cluster AWGR-based switches (EAS), and N ToRs within each cluster connected by intra-cluster AWGR-based

TABLE 3. Hybrid switches comparison.

Ref.	Structure	Power (P_{π})	Bandwidth	Crosstalk	Extinction Ratio	Loss	Footprint
[24]	16×16 switch based on Dual ring assisted MZI (E-O, T-O)	18.7 mW	0.33 nm	<-20.5 dB	-	10.6 ± 1.7 dB	N/A
[50]	2×2 CDC MZI (T-O)	-	3 nm	<-28 dB	N/A	< 2.5 dB	N/A
[25]	2×2 16 channel CDC MZI (T-O)	3.8 -15.6 mW	4.2 nm	-	15 dB	< 5.4 dB	N/A
[49]	2×2 FSR free MRR with CDCs	N/A	0.28 nm	N/A	16.5 dB	1.6 dB	1963 μm^2
[51]	2×2 Double ring assisted MZI (E-O)	0.69 - 2.31 mW	0.48 nm	<-20 dB	-	1.6 dB	N/A
[106]	4×4 Double ring assisted MZI (E-O)	1.38 mW	35 GHz	-18.4 dB	-	4-5.8 dB	N/A

switches (IAS). The network supports K servers interconnected within each ToR switch. Packet forwarding in this network relies on a fixed cyclical-permutation-based routing matrix between the input and output ports of the AWGR. However, some potential problems with this approach could include limited flexibility in routing, as the packet forwarding relies on a fixed routing matrix and potential congestion or bottlenecks within the intra-cluster and inter-cluster switches. Further, the complexity of the architecture may introduce challenges in terms of fabrication, cost, and management of the network. The internal structure of the TOR switch is demonstrated in Figure 8.

In [122] a structure called Flex-LIONS is described that is based on cooperation between AWGRs and MRRs. Flex-LIONS is a reconfigurable optical interconnection architecture that utilizes an N-port AWGR (Arrayed Waveguide Grating Router) at its core to provide all-to-all connectivity. It employs b MRR (Microring Resonator) add-drop filters at each AWGR input/output port to drop certain wavelength channels. By tuning the MRR add-drop filters and using an $N \times N$ strictly non-blocking multi-wavelength MRR-based crossbar switch, the bandwidth between input and output ports can be effectively increased by a factor of b. The Flex-LIONS device is fabricated on a multi-layer platform, consisting of a silicon layer with MRR add-drop filters and a multi-wavelength MRR crossbar switch, and a silicon nitride (Si_3N_4) layer containing an 8×8 cyclic AWGR. These layers are vertically interfaced through inverse-tapered evanescent couplers. Figure 9 shows the schematic of Flex-LIONS proposed in [122].

Recently in [123] a monolithic SiPh 32×32 Thin-CLOS array waveguide grating router is proposed which uses four 16-port AWGRs to create an all-to-all interconnection with adjacent channel crosstalk of less than -20 dB.

2) SOA-MZI SWITCHES

There are various SOA-based switch structures, including Terahertz Optical Asymmetric Demultiplexer (TOAD), Colliding Pulse Mach-Zehnder (CPMZ), and Symmetric Mach-Zehnder (SMZ) where they all utilize the properties of SOAs in conjunction with Mach-Zehnder Interferometer

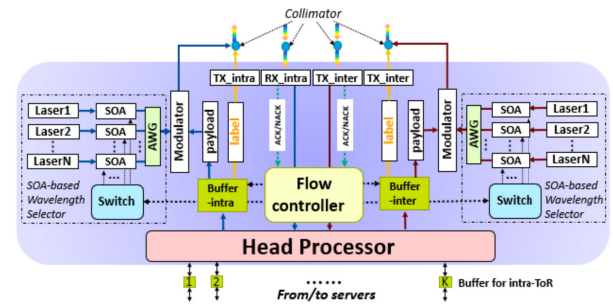


FIGURE 8. Schematic of the TOR switch reprinted form [118]. ©2022 Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

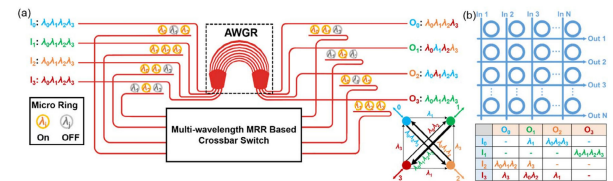


FIGURE 9. Schematic of Flex-LIONS reprinted from [122]. ©2019 IEEE.

(MZI) structures [124]. The structure of these switches can be divided into two categories linear (gating) or nonlinear (wavelength conversion) based on their operational tasks. Control pulses create a differential phase shift in switches based on the Mach-Zehnder design [124], [125]. Figure 10 demonstrates the structure of the SMZ and CPMZ. The MZI component provides high-precision switching capabilities while the SOA component adds gain, enabling the switch to amplify the optical signal as well. This hybrid design allows for increased stability, lower power consumption, and reduced size compared to conventional optical switches. Hybrid MZI and SOA optical switches are widely used in data centers, optical networks, and various other applications where high-speed and efficient optical signal routing is required.

In [126] the paper reports the first successful demonstration of a hybrid dilated 2×2 modular optical switch

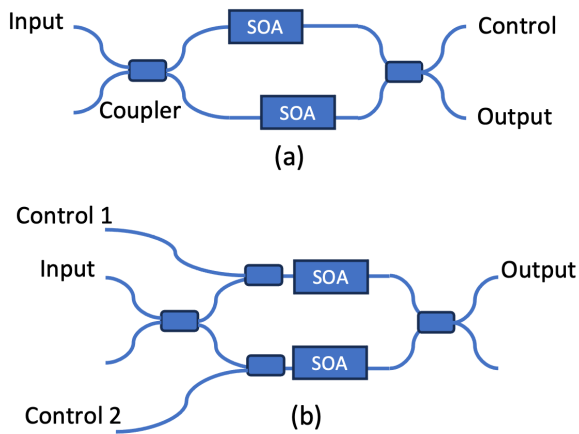


FIGURE 10. Schematic of (a) CPMZ and (b) SMZ redrawn from [124].

using Mach-Zehnder modulators and short SOAs for low-loss switching and increased extinction and gain. The switch achieved an excellent cross-talk/extinction ratio of 40dB and a data-modulated signal-to-noise ratio of 44dB in 0.1nm bandwidth. With a fast switching time of 3ns and low subsystem penalties of less than 0.1dB, this hybrid switch has the potential for building an 8×8 port switch with 14dB input power dynamic range and low subsystem penalties of less than 0.5 dB.

In [127] authors demonstrated a 4×4 Hybrid MZI-SOA InGaAs/InP photonic integrated switch which is implemented by a combination of active and passive integration of MZI and SOA optical switch, MZI is used as the main switching element and short, switchable SOAs are utilized to compensate for loss and suppress crosstalk. The switch employs a dilated Benes architecture to reduce crosstalk and offer rearrangeable non-blocking connections.

In summary, SOA-MZI-based photonic switches are innovative devices that enable efficient signal processing for both amplitude and phase-modulated signals, crucial for modern optical networks. Capable of handling various signal formats like On-Off Keying (OOK) and Phase Shift Keying (PSK) at up to 40 Gb/s, they adapt seamlessly to network changes, improving system efficiency and reducing power consumption by operating directly in the optical domain. The main challenge with SOA-MZIs is the carrier lifetime in SOAs, affecting processing speed and efficiency [128]. Nonetheless, continuous advancements are expected to mitigate these limitations, promising enhanced performance for future optical communications.

C. III-V/ Si HYBRID SWITCHES

Hybrid III-V/Si switches are a type of semiconductor switch that combines the benefits of III-V materials and silicon to create high-performance optoelectronic devices. These switches utilize the unique optoelectronic properties of III-V materials, such as high light absorption and emission efficiency, while leveraging the mature silicon

processing technology. This combination results in devices with improved efficiency, speed, and reliability compared to traditional silicon-only switches, making them suitable for a wide range of applications in telecommunications, data centers, and other optical communication systems. The improvement of wafer-bonding technology has given rise to a new category of integrated devices, where III-V materials are attached to silicon wafers either directly [129] or with the use of polymer adhesives [130]. This combines the best features of both materials, allowing for mass production, and presents a possibility for the development of compact, energy-efficient, and cost-effective switch fabrics that meet specific performance standards.

Previously in [131] the integration of III-V materials with the silicon-on-insulator (SOI) platform through wafer-bonding has been explored as a way to improve the E-O effect for phase manipulation. To date, most attempts to add III-V gain blocks to silicon switch circuits have utilized flip-chip bonding, which requires precise material coupling and low reflection at both facets and may detract from the cost-effectiveness of silicon fabrication. The standalone hybrid SOA device was first demonstrated by bonding an III-V layer onto a silicon waveguide to confine the optical mode laterally [132]. This design offers less complexity in inter-layer coupling but limited modal gain and requires a tapered structure to increase coupling efficiency and minimize reflection. Variations in the silicon waveguide width allow for a trade-off between optical gain and saturation power, but long SOAs are needed due to the restricted modal gain. Another design proposes having the optical mode reside in the III-V waveguide to achieve maximal modal gain and uses adiabatic mode transformers for hybrid lasers [133], which can be as efficient as a monolithic InP laser diode. However, this design requires careful examination of the adiabatic coupler to ensure high-efficiency and low-reflection power transfer and may lead to a large footprint due to the complexity of the adiabatic mode transformers. In addition, the complexity of adiabatic mode transformers could affect device yield and lead to a large footprint. The energy efficiency of III-V/Si hybrid switches largely depends on the coupling efficiency between the two materials, which remains a challenge [110], [134].

A 2×2 optical switch based on the III-V/Si hybrid MOS phase shifter introduced in [17]. The significance of this work lies in the demonstration of an ultra-power-efficient 2×2 Si MZI optical switch that utilizes III-V/Si hybrid metal-oxide-semiconductor (MOS) phase shifters. The hybrid MOS capacitor exhibits negligible gate leakage current, resulting in a power consumption of only 0.18 nW for switching, significantly smaller than that of a Si T-O phase shifter. Besides, the demonstrated switching time of less than 20 ns is an improvement over many existing technologies. Fig. 11 demonstrates the proposed structure in [17].

In [135] a similar approach was used where an III-V/Si hybrid MOS optical phase shifter with an ultrathin InP

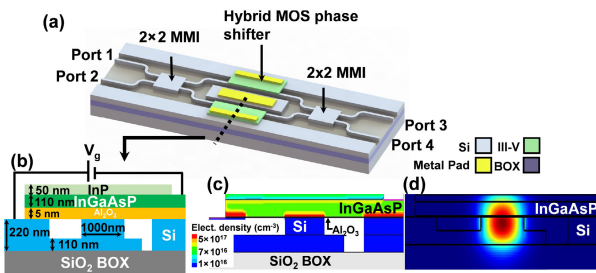


FIGURE 11. The hybrid MOS phased shifter MZI switch introduced in [17]. Figure reprinted from Q. Li, J.-H. Han, C. P. Ho, S. Takagi, and M. Takenaka, 'Ultra-power-efficient 2×2 Si Mach-Zehnder interferometer optical switch based on III-V/Si hybrid MOS phase shifter,' *Optics Express*, vol. 26, no. 26, pp. 35003–35012, 2018. Licensed under a Creative Commons Attribution (CC BY) License.

membrane to design an ultralow-power Si microring resonator switch. Electron accumulation at the InP MOS interface enables efficient optical switching with a minimal power consumption of only 0.3 pW, allowing for extensive integration without the issue of thermal interference.

In [136] a new type of optical switch based on a microring resonator (MRR) switch was introduced. It uses an III-V/Si hybrid metal-oxide-semiconductor (MOS) optical phase shifter with an ultra-thin InP membrane. The switch is designed to reduce the insertion loss of the phase shifter by reducing the thickness of the InP membrane, resulting in an MRR switch with a high-quality factor (Q-factor). By carefully optimizing the device structure through numerical analysis, we were able to demonstrate a successful proof-of-concept MRR optical switch. This switch requires only 0.3 pW power for switching, making it ideal for power-efficient, thermal-crosstalk-free, Si-based photonic integrated circuits that use WDM.

These approaches utilize an ultralow-power optical phase shifter to replace power-consuming thermo-optic (TO) phase shifters. The hybrid integration with an ultrathin III-V membrane enables the monolithic integration of photodetectors and optical power monitors, making it promising for applications in communication, computing, and sensing. The main benefits include reduced power consumption and the ability to combine multiple functions in a single device [17], [135], [136], [137].

The challenges of monolithic integration of III-V and Si materials through the direct wafer bonding process are primarily associated with material incompatibilities and the subsequent impact on device fabrication and performance. Cross-contamination risk, wafer size and cost, and temperature constraints for III-V growth are some of the challenges of monolithic integration of III-V and Si materials [138].

D. MEMS

Opto-mechanical switches, also referred to as Micro-Electro-Mechanical Systems (MEMS) switches, employ the movement of parts to steer light from one port to another. These switches make use of optical components, like prisms or

micro-mirrors, to guide the optical signal from one fiber to the target output fiber [110]. MEMS switches are the most common and well-established free-space switching technology. They are created by either bulk micro-machining, where the mechanical structures are directly etched into a silicon substrate, or surface micro-machining, which involves depositing, patterning, and removing epitaxial layers such as polysilicon, silicon nitride, and silicon oxide. MEMS switches come in two configurations based on signal propagation free space and waveguide-based switches. In this section, we will have a review of both of these structures and their work principles.

1) FREESPACE SWITCHES

Two-dimensional (2-D) and three-dimensional (3-D). The key difference between these two configurations is the movement of the switching mechanism. 2-D switches have a flat mirror that rotates on a hinge, while 3-D switches have a moving mirror that can be translated more complexly to redirect light. Electrostatic drivers are the most frequently used because they are low-power and easy to control, although the typical operating voltage is high, often reaching 100–150 volts [110], [139].

In a 2-D MEMS optical switch configuration, 2×2 -switch components are organized in a grid-like pattern with M rows and M columns. Each of these elements is a bi-stable micromirror capable of adopting either a bar or cross state. However, the achievable maximum port count for 2-D MEMS switches is restricted due to fabrication limitations, and their switching time is comparatively lengthy (on the order of milliseconds) as a result of mechanical actuation [110], [139], [140], [141].

Conversely, 3-D MEMS optical switches employ micro-mirrors for directing beams. This design can support high-port-count switching, but there is a trade-off between the port count and switching time since the number of beam-steering states is proportional to the square of the switch port count. The highest port count reported is 1100 [142], with switching times of approximately 5 ms for a 256-port switch and 150 μ s for a 61-port experimental switch [140].

2-D MEMS switches are MEMS-based devices that are used to switch optical signals in optical communication networks. They use a 2-D bi-stable micro-mirror to redirect incoming light from one optical fiber to another. However, 2-D MEMS structures are not easily scaleable because for an $N \times N$ switch we will need N^2 micro-mirrors, and the fabrication is bottlenecked by the precision in the alignment of the micro-mirrors [110], [143] since larger mirrors are required for the larger port count switches [143].

A demonstration of a 2-D MEMS switch is shown in Fig. 12. The light is either reflected or passed through uninterrupted. The main drawback of a 2-D MEMS switch is the non-uniform insertion loss. Because the distance between the light path toward outputs varies for different port combinations [110].

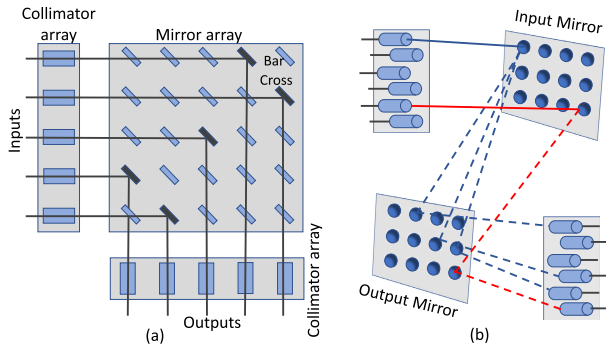


FIGURE 12. Diagrams of MEMS optical switches. (a) A 2D MEMS switch configuration with collimator arrays for input and output signals. The mirror array allows for either "bar" or "cross" connections, directing optical signals between inputs and outputs. (b) A 3D MEMS switch, where input and output mirrors are placed at different planes. Optical signals are routed through multiple mirror layers, providing greater flexibility and scalability for complex switching between inputs and outputs. The red and blue dashed lines represent different optical paths within the switch. Figure redrawn from [145].

3-D MEMS switches. The typical design of a 3-D optical switch involves utilizing two arrays of mirrors, each set to correspond with a group of collimated, parallel, fibers for either input or output. This setup requires the use of $2N$ mirrors in a switch with N ports, which is a significantly smaller number compared to the 2-D architecture [110].

This architecture can be expanded to accommodate more ports while maintaining a consistent level of insertion loss. Its main benefit is that the variations in the path of light between different ports during switching are less influenced by the number of ports. Additionally, this design can be improved further by using a Fourier transform lens between the micro-mirror arrays, which results in lower maximum angle requirements, smaller micro-mirror size, improved tolerance in micro-mirror curvature, and reduced switch crosstalk [21], [144].

The design of a 3-D MEMS optical cross-connect switch is challenging due to the need to harmoniously balance optical design, MEMS design, and mechanical design in order to attain optimal optical performance. Key factors in choosing a mirror for this type of switch include its size and fill factor, deflection angle, reflectivity, resonant frequency, driving voltage, flatness, curvature, and control system. A 3-D MEMS is demonstrated in Fig. 12.

2) MEMS WAVEGUIDE SWITCHES

Waveguide MEMS switches operate by keeping the optical signal contained within waveguides, without allowing it to disperse into free space. The switch accomplishes the transfer of the optical signal from one waveguide to another by precisely positioning a waveguide or coupler, facilitating the coupling between them and enabling the signal to move from the original waveguide to the target waveguide [141]. In the late 1970s, the idea of moving fibers for optical switching [146], [147] with the advent of MEMS technology the moving fibers were replaced with waveguides. The

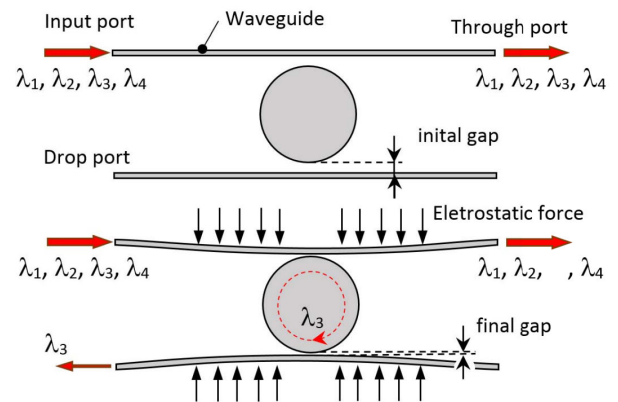


FIGURE 13. The basic work principle of MEMS-based microdisk switches [141]. Reprinted from [141]. Copyright ©2019, IEEE.

main advantages of MEMS-based moving-fiber/waveguide switches include low insertion loss across a broad wavelength range and faster switching times (from a few milliseconds to under 1 ms) compared to traditional optomechanical counterparts [141].

Various methods can be employed to implement MEMS waveguide switches. One such method is the MEMS-based micro-disc switch, as depicted in Fig. 13. This approach uses the coupling coefficient between the micro-disc and the waveguide, which varies exponentially with the gap spacing [20]. The spacing between the micro-disc and the waveguide is controlled by electrostatic force. When the waveguides come close enough, only wavelengths that match the resonant wavelength of the micro-disc can pass through it. In Fig. 13, the micro-disc is tuned to wavelength λ_3 , which, if desired, can be directed to the drop port. In [148] an add-drop ring resonator that enables independent tuning of round-trip phase and coupling using low-power MEMS actuation (see Fig. 14). At a wavelength of 1540 nm and a maximum voltage of 40 V, the phase shifters can tune the resonance wavelength by 0.15 nm. The resonance exhibits a passive quality factor of 29,000, which can be increased to nearly 50,000 with actuation. Moreover, the MEMS rings are vacuum-sealed individually on a wafer scale, providing reliable, long-term environmental protection. To minimize round-trip loss typically caused by discrete anchors, the ring resonator waveguide is designed as a continuous single-side-suspended rib waveguide. In [149] authors present a compact add-drop filter based on a MEMS-actuated, vertically movable ring resonator integrated within IMEC's iSiPP50G SiPh platform. The proposed filter offers a through port linewidth of about 1 nm (124.37 GHz) at 1557.1 nm, retaining a port extinction of 20 dB and a port isolation of over 50 dB under 27 V of the actuation voltage.

In [150] the platform integrates silicon nitride (Si_3N_4) based optical waveguides and overcomes the challenges associated with 3-D MEMS, such as complex structures and tight alignment tolerances. The proposed structure utilizes

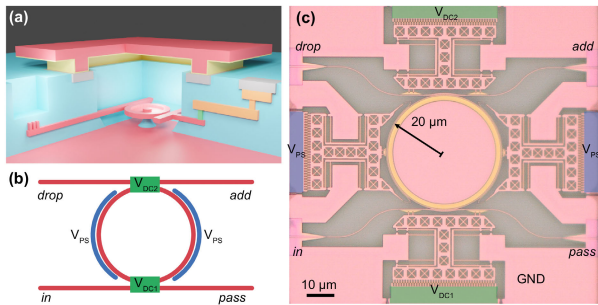


FIGURE 14. The add-drop Vacuum-sealed ring resonators from [148] (a) 3-D demonstration of the structure. (b) Microscopic image of the manufactured device utilizing comb-drive MEMS actuators for the independent adjustment of input coupling, output coupling, and round-trip phase. (c) Top-view optical micrograph of the fabricated MEMS-enabled add-drop ring resonator. Electrical contacts and anchor points are visible around the MEMS cavity perimeter, enabling actuation and integration with standard photonic waveguides. Reprinted from [148]. Licensed under a Creative Commons Attribution (CC BY) License.

simple parallel plate actuators to achieve lateral displacement and efficient switching between different input and output waveguides. The operating voltage of this switch is 65 V and a 2×2 switch can be accomplished with a footprint of 1 mm by 1 mm. Work in [151] presents a novel optical switch based on a laterally rotating micromachined platform integrated under optical waveguides. The platform controls the in-plane propagation direction of light within a die to select one of the multiple outputs. Designed to exhibit low constant optical losses, the platform is electrostatically actuated using an optimized circular comb drive, achieving an angular motion of $\pm 9.5^\circ$ with 180 V. A gap-closing mechanism minimizes optical losses, while a latch structure ensures a 0.25° resolution over the entire motion range. Integrated with silicon nitride waveguides, the platform was used to create a crossbar switch with preliminary optical measurements reporting 14.8 dB loss in the bar state and 12.2 dB in the cross state.

Recent advancements in MEMS technology have significantly enhanced traditional SiPh and other technologies by addressing key limitations such as power consumption, component size, and integration density.

3) RECENT ADVANCEMENTS IN MEMS OPTICAL SWITCHING DEVICES AND THEIR IMPACT

MEMS technology has seen significant advancements in recent years, enhancing the performance and scalability of optical switching devices. These developments address critical challenges such as power consumption, footprint, switching speed, and integration density, thereby enabling their deployment in cutting-edge applications.

One major advancement is the development of tunable phase shifters and couplers with negligible power consumption ($<1 \mu\text{W}$ per device) and compact footprints, as small as $100 \times 45 \mu\text{m}^2$, enabling broadband performance. These components are crucial for programmable photonics and telecommunications, where low power consumption and

high efficiency are essential [19]. Recent developments also include bistable MEMS phase shifters with non-volatile actuators, achieving phase shifts over 2π with optical losses as low as 0.03 dB [152].

Integration of MEMS actuators into SiPh platforms has enabled very large-scale integration (VLSI) by overcoming the limitations of traditional thermo-optic phase shifters (high power consumption and thermal cross-talk) and plasma-dispersion-based modulators (large footprints). This advancement provides precise and fast reconfiguration capabilities with low insertion losses, which is ideal for photonic integrated circuits (PICs) in applications like data centers [19], [20].

High-speed MEMS-based optical switches now achieve extinction ratios exceeding 23 dB and MHz-level reconfiguration speeds, offering significant advantages for optical communication networks and data centers. These switches enable energy-efficient and high-speed signal routing, making them competitive with other emerging technologies [19], [150].

Moreover, MEMS technology has introduced wafer-level hermetic sealing, which protects sensitive optical components from environmental factors and ensures long-term reliability. This innovation enhances the robustness and industrial viability of MEMS-enhanced optical devices [148].

Recent efforts have also focused on MEMS-based micro-disc switches and waveguide MEMS switches, which leverage parallel plate actuators for precise lateral displacement, achieving sub-millisecond switching speeds and compact designs. These developments represent a clear advancement over traditional free-space MEMS systems, which were often limited by larger footprints and slower switching times [141].

E. SILICON NITRIDE AND POLYMER-BASED SWITCHES

Silicon nitride (Si_3N_4) waveguides have been a topic of interest since the late 1970s, with the early motivation to develop photonic circuits that use silicon substrates and silicon processing technologies while operating at wavelengths where silicon absorbs [153], [154]. From the initial fabrication of thin-film slab waveguides in the late 1970s and early 1980s [155], [156], researchers focused on reducing losses by optimizing material density and minimizing waveguide roughness. In the 1990s [157], the first Si_3N_4 PICs were fabricated, and Silicon oxynitride devices were developed, advancing the nitrides' capabilities.

The commercialization of Si_3N_4 waveguides began in the 1980s and early 1990s, with various foundry platforms evolving, including LioniX's TriPleX, Ligentec's Damascene, IMEC's BioPIX, and IMB-CNM processes. Advances in thick nitride waveguide technology continue to reduce losses, and CMOS compatibility illustrates the benefits of foundry processes [153].

Silicon nitride technology has been increasingly utilized in the design of optical switches due to its unique combination of properties, such as high thermal stability, high resistance

to chemical corrosion, and good electrical insulation. These properties make it an attractive material for use in optical communication systems, particularly for the design of waveguide crossings [158], MZI switches [159], and MRR switches [160].

The use of silicon nitride in optical switch design offers several benefits. Its high thermal stability allows for improved performance in high-temperature environments, making it ideal for use in applications such as data centers and high-performance computing systems. Its good electrical insulation properties make it well-suited for use in optical communication systems where electrical isolation is required [153], [155], [157], [160], [161].

There's ongoing research on the design of optical switches on the Si_3N_4 platform; here we briefly mention some of these works and their achievements over the past few years. In [162] authors proposed a T-O Switch based on a silicon Nitride waveguide buried in polymer to utilize the T-O properties of the polymer to achieve different functionalities such as polarization-independent path switching, beam splitting, and polarization beam splitting. In the fabricated device designed for polarization-independent path switching, the achieved crosstalk suppression has been found to exceed 10 dB for the TE mode and over 20 dB for the TM mode within the wavelength range of 1500 nm to 1620 nm (see Figure 15).

In [163], researchers demonstrated an O-band MZI based on an N-rich silicon nitride platform and Phase change material (PCM) tuning, which is explained in the tuning mechanism section, targeting potential optical communication applications. The device's functionality relies on controlling the waveguide's losses using a PCM cell, which can switch between amorphous (low-loss) and crystalline (high-loss) states. The MZI optical building block achieved an extinction ratio of up to 11 dB between the amorphous (ON) and crystalline (OFF) states. The insertion loss for the MZI structure was as high as 0.87 dB/m in the OFF state and as low as 0.064 dB/m in the ON state for TM polarization.

In [159], a 2×2 silicon nitride MZI T-O switch with a 1mm-length phase shifter was fabricated on a bulk silicon wafer using a CMOS-compatible process. The device designed for 1310 nm wavelength demonstrated approximately 26.8 dB crosstalk, 0.23 dB insertion loss, and $17.25 \mu\text{s}$ switching speed at around 1325.8 nm. Similarly, the device for the 1550 nm wavelength exhibited about 24.7 dB crosstalk, 0.48 dB insertion loss, and $17.48 \mu\text{s}$ switching speed at roughly 1550.8 nm. The measured operating power was 55.6 mW and 64.4 mW at around 1325.8 nm and 1550.8 nm wavelengths, respectively.

In [164], a 32×32 optical switch fabric was demonstrated based on a switch-and-select (S&S) topology. The S&S structure is a strictly non-blocking switching topology with fully suppressed 1st-order crosstalk. However, numerous in-path waveguide crossings significantly degrade path-dependent insertion loss. By utilizing a dual-layer Si_3N_4 -on-SOI platform, the researchers created a 32×32 MZI switch chip

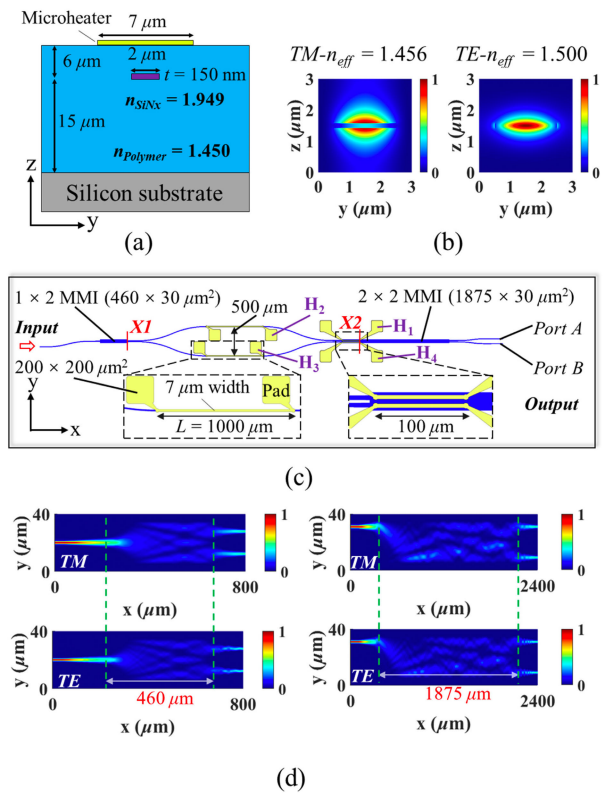


FIGURE 15. Demonstration of the structure proposed in [162] (a) cross-sectional view of the waveguide buried in polymer material. (b) Simulated TE and TM mode in the waveguide. (c) schematic of the T-O switch. (d) Simulated results of the 1×2 and 2×2 MMI structure for TE and TM mode. ©2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

with low path-dependent insertion loss, low crosstalk, and low power consumption.

In another work, [160], the authors proposed and experimentally demonstrated a high-performance all-optical switch using graphene-on-silicon nitride (Si_3N_4) in a MZI configuration. The device operates by having the graphene on the Si_3N_4 waveguide absorb part of the pump light power, generating heat that changes the waveguide's refractive index due to the thermo-optic effect. This process allows for all-optical switching by tuning the phase of the probe light in the Si_3N_4 arm with graphene on top.

Authors in [165] present a dual-mode 2×2 T-O switch based on an MZI with symmetric directional couplers, using polymer waveguides. The experimental device has a total length of 30 mm, a switching power of about 9.0 mW, and a switching time of around 1.30 ms.

In [166] the authors proposed a multiport T-O switch based on polymer waveguides is introduced. In the proposed architecture an array of polymer waveguides and by applying heat through the heaters the switch can transfer light in two up-and-down or isolation transfer fashions. The proposed

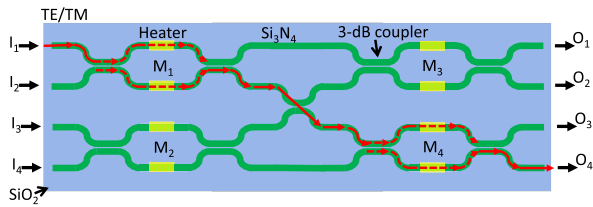


FIGURE 16. Schematic of the 4×4 optical switch as described in [161], redrawn from the original paper. The switch consists of a series of MZIs (M_1, M_2, M_3, M_4), heaters for thermal tuning, and 3-dB couplers for signal splitting and combining. The optical paths for both TE and TM modes are shown, with input ports (I_1 – I_4) and output ports (O_1 – O_4) indicated. The structure is based on a Si_3N_4 platform, with SiO_2 as the cladding material.

structure has a power consumption of 33.8 mW and a coupling length of $1331.6 \mu\text{m}$. The proposed structure can be scaled into $M \times 1$ and a $1 \times N$ switch network structures.

Authors in [161] proposed a 4×4 optical switch based on Si_3N_4 structure (see Fig. 16). The switch is composed of four 2×2 switching elements cascaded with a power consumption of less than 130 mW for a single switching operation. The measured rise/fall time of the switch upon electrical tuning is about 24.5/30.5 μs .

Silicon nitride stands out for its wide wavelength transparency (400–2350 nm), ultra-low propagation losses ($<1 \text{ dB/m}$), and excellent high optical power handling capabilities. This versatile platform, compatible with CMOS processing, enables both linear and nonlinear optical functions, making it suitable for a broad range of applications from visible to mid-infrared spectra [159], [160], [164], [167]. Despite its advantages, challenges such as material stress and increased fabrication complexity, alongside potential cost implications, need careful management. Nonetheless, Si_3N_4 's integration into photonic integrated circuits promises significant advancements in optical communication, sensing, and computing technologies.

F. LCOS

Liquid crystal technology has been around for many years and initially was introduced for image and video display applications. This technology is based on a combination of light-modulation characteristics of liquid crystals and the benefits of CMOS technology, using specific LCOS assembly techniques [168]. LCOS devices have various applications including WSS as discussed in [169], [170]. LCOS optical switches are a type of optical switch that utilizes liquid crystal material and silicon-based technology to control the flow of optical signals. In LCOS optical switches, the liquid crystal material is sandwiched between two glass plates and illuminated by a light source. By applying an electric field, the liquid crystal molecules can be aligned to allow or block the flow of light, enabling the switch to direct optical signals in a controlled manner [168].

Recently, [171] proposed and investigated a 3×3 LCOS-Based switch. This WSS is designed for the C-band, with an insertion loss of less than 13.2 dB, a spectral

resolution of 1 GHz, and a bandwidth tuning range of 16.25 to 3750 GHz. Since the LCOS switches have large footprints recent works such as [172] are focused on proposing solutions to increase the steering angle of LCOS devices using techniques such as integration with dielectric metasurfaces.

G. CHALLENGES IN OPTICAL SWITCHING DEVICES

Designing an optical switch is a complex task that requires addressing several technical challenges. One major challenge is maintaining low loss levels while switching the optical signal. Another challenge is the insertion loss, which is caused by factors such as the physical size and absorption and can degrade the overall performance of the switch. Cross-talk between the channels of an optical switch is also a concern, as it can lead to signal degradation. The wavelength dependency of optical switches is another challenge, as they need to be able to operate over a wide range of wavelengths. Optical switches can also be polarization-sensitive, which can cause problems with polarization-maintaining fibers. Reliability and durability are also important factors to consider in optical switch design, as these components need to be able to withstand environmental conditions and maintain their performance over time. Finally, the cost can also be a challenge, as optical switches are typically complex and expensive components that may not be feasible for mass-market applications. In this section, we will review some of the most important challenges that currently exist in the design of optical switches.

1) POLARIZATION DEPENDENCY

Polarization dependency can play an important role in the design of optical switches since it can limit the application of the switch. Some photonic devices are designed for single polarization such as SOA-gated switches that are designed for TE polarization, however, devices can be designed to be polarization insensitive to exhibit the same performance for TE and TM modes [110]. In other words, if the loss of the optical switch is not the same for different polarization (polarization dependant loss (PDL)) it's said that the switch is polarization dependant it's ideal to have a small PDL [173].

In the design of MZI switches the two power splitters in the MZI structure impose the Polarization dependency on the switch structure, as a result, many works focus on the design of engineered MMI and directional couplers to be polarization independent [161], [162], [166], [174], [175], [176]. One of the key parameters in the design of optical power splitters is the etch depth which can impact the polarization dependency of the devices optimal etch depth can help the polarization-independent operation of the couplers however slight errors in the fabrication and etching process can lead to polarization dependence operation [107], [175]. As another example of polarization-dependent devices, we can mention CDCs where all the proposed devices can only operate in TE polarization [45], [48], [93], [177]. Devices formed with deep

etching depth, which are also known as high-mesa structures, have lower cross-talk characteristics, and modes are more strongly confined compared to the middle mesa structures [107], [178].

2) PERFORMANCE PARAMETERS

The design of optical switches presents several challenges that must be overcome to achieve high performance and reliability. One of the major challenges is the small footprint of optical switches, which requires the use of compact and high-density optical components [12], [173]. Crosstalk, or the unwanted interference between adjacent optical channels, is one of the important issues that must be minimized to ensure the accuracy and integrity of the data being transmitted. Crosstalk is the proportion of power at a particular output that comes from a specified input, compared to the power from all other inputs [12], [173].

The data rate and bandwidth requirements for modern optical switches are also increasing, requiring the switch to be capable of handling high-speed optical signals with low loss and high extinction ratio. Additionally, the insertion loss of the switch, which is the amount of light lost when the switch is engaged, must be kept low to minimize the impact on the signal quality and transmission distance. These challenges require a careful and comprehensive approach to the design and engineering of optical switches, taking into account the trade-offs between performance, size, cost, and reliability [11], [12]. In broadband applications, the bandwidth limitation of some of the optical switch designs can be hindering and the design of wideband optical switches has gained much attention [44], [47], [93], [94], [179].

Many research papers have focused on the improvement of the optical switches' performance such as push-pull control or nested structures and broadband couplers. For instance, a balanced tri-state nested MZI structure was introduced [180] where the blocking state could help the cross-talk performance of the MZI by more than 50 dB in the worst output crosstalk. In addition, [181] reported the performance of 2×2 , 4×4 and 8×8 SiPh thermal banyan switches and experimentally evaluated the performance of the switch. These multimode interference-based switches demonstrated low loss, low crosstalk, low power penalty, and a bit error rate below 10^{-10} during payload data transmission [181]. In [72] researchers achieved a smaller footprint for MZI switch by loading an asymmetric doped MZI structure and an integrated heat sink into the optical switch. In the proposed structure, the distance between both phase shifters was reduced to $2 \mu\text{m}$, resulting in miniaturization.

Another aspect of designing optical switching elements is the footprint or the amount of real state each switching element will occupy on the silicon wafer. The footprint is a crucial aspect in the design of optical switches as it determines the physical size of the switch, which directly impacts its integration into photonic integrated circuits or

optical networks. A smaller footprint enables a higher density of optical switches to be integrated, resulting in more efficient use of space and greater functionality. Minimizing the footprint of optical switches is an important goal in the design process, and its optimization is essential for the development of high-speed and compact optical networks [14], [42], [113].

Insertion loss is a key performance metric in optical switches as it measures the amount of optical power lost as light passes through the switch. It is defined as the difference between the optical power before and after the switch, expressed in decibels (dB). A lower insertion loss indicates a more efficient switch, as more of the optical power is transmitted without being absorbed or scattered. The insertion loss of an optical switch is influenced by various factors such as the material used, the size and shape of the switch, and the fabrication process. Therefore, reducing the insertion loss is a critical consideration in the design of optical switches, as it affects the overall transmission quality and efficiency of optical communication systems [113], [173].

III. TUNING MECHANISMS

Optical switch tuning mechanisms refer to the methods used to control the switching of optical signals in optical switches. This typically involves adjusting the parameters of the switch such as its optical power, wavelength, polarization, etc. to control the flow of optical signals through it. Some common tuning mechanisms include electrical tuning, thermal tuning, mechanical tuning, and acousto-optic tuning. These mechanisms allow for precise and rapid control of the optical signal routing, making optical switches useful in optical communication networks, data centers, and other optical systems.

This chapter provides an overview of the different methods used to control the switching of optical signals in optical switches. It discusses various tuning mechanisms, including electrical tuning, thermal tuning, mechanical tuning, acousto-optic tuning, phase change materials, and their applications in different optical systems. By understanding these tuning mechanisms, readers will gain a comprehensive understanding of how optical switches can be tuned to route optical signals efficiently and effectively in optical communication networks, data centers, and other optical systems.

A. T-O AND E-O

One of the most well-known tuning mechanisms of optical switch control is based on optical interference. Optical interference actuated switches are optical switches that use the interference of light waves to control the flow of optical signals which was discussed in the previous section such as MZI switches. These switches work by creating an optical interference pattern in the switch that can be used to direct incoming optical signals in different directions. The interference pattern is generated by combining two or more light waves of the same frequency and direction but with different phases [11], [12], [113]. The interference pattern can be adjusted by changing the relative phase of the light

waves, which allows for precise control over the flow of optical signals. To create a phase difference the most common way would be changing the effective refractive index of the waveguide by deploying phase shifters. A phase shifter can be subdivided into a T-O switch, which uses heat to change the index of refraction in silicon, and an E-O switch, which uses a p-i-n junction diode which will be discussed in this section.

1) THERMO-OPTIC

In the T-O approach, the flow control of optical signals in optical switches is achieved by adjusting the refractive index of the switch's optical material through heating. The refractive index of material changes with temperature, and by heating a section of the switch, the refractive index of that section can be changed. This allows the optical phase of the incoming signals to be changed, which is a key aspect of the functionality of T-O switches.

The magnitude of the change in refractive index is proportional to the interaction length (the distance the light travels through the material) and the wavelength of the light. Optical confinement, on the other hand, refers to the extent to which the light is confined within a material, and can also influence the magnitude of the T-O effect. In general, materials with higher optical confinement and longer interaction lengths will have a larger T-O effect for a given wavelength of light [12].

Equation 1 from [12] shows that the phase change of the light is proportional to the product of the wavelength of the light, the change in the refractive index of the material, and the interaction length. The mode confinement factor is included to account for the influence of optical confinement on the magnitude of the T-O effect.

$$\Delta\phi(\lambda, T) = \frac{2\pi}{\lambda} \Delta n(T) \Gamma L \quad (1)$$

where in this equation the Γ is the confinement factor of the waveguide and $\Delta n(T)$ is the change of index due to the temperature change.

For silicon waveguides, the T-O coefficient is $1.86 \times 10^{-4}/K$ meaning that for every 1 Kelvin increase in temperature, the refractive index of the silicon waveguide changes by 1.86×10^{-4} . This coefficient is 2.35×10^{-4} (GaAs), 2.01×10^{-4} (InP) for III-V materials [182], [183], [184]. The T-O coefficient is a measure of the change in the refractive index of a material with temperature. In the case of silicon waveguides, this coefficient is used to understand and predict the changes in the waveguide's optical properties due to temperature changes, which can be important in the design and optimization of optical devices. Higher T-O coefficients can allow for a larger change in the refractive index of the material with temperature, which can result in a larger phase change in the optical signal [12]. This can, in theory, allow for switching with less power. However, in practical optical switching systems, a higher T-O coefficient can also result in larger thermal disturbances in the optical signal, which can cause instability and reduce the system's

reliability. Therefore, while a higher T-O coefficient may allow for switching with less power, the trade-off in terms of increased thermal disturbance and potential instability makes it challenging to use materials with high T-O coefficients in practical optical switching systems.

TO switches usually have power consumption in the range of $10\text{--}30 \text{ mW}/\pi$ which can be different for various designs and heater materials. The response time is in the order of microseconds [12]. The T-O switch is ideal for applications requiring millisecond to microsecond response time, as it can switch on and off within the time frame of microseconds to milliseconds. This makes it a great option for inter-data center connections and the switching needs in telecommunication network nodes.

There are various methods for the implementation of TO heaters such as front-end and back-end of the line (FEOL and BEOL) methods as discussed in [12] among these two methods BEOL is the most common method. In the resistive heating method, which is a common approach, a resistive material, such as gold or titanium nitride, tungsten [12], [185], [186] is positioned above the waveguide with a distance of 1 to $2 \mu\text{m}$. As the distance between the heaters and the waveguide increases the response time of the switch can also increase which is one of the disadvantages of the T-O tuning mechanism [12]. Another approach is the doping of the silicon which can change its electrical resistivity, allowing the film to act as a resistive heater. Using the waveguide as a resistor by lightly doping is also an option, however, it will cause an increased loss in the waveguide [11], [12].

There exist methods to enhance the effectiveness of T-O phase modulation, one solution could be to incorporate a deeply carved cavity that encompasses both the waveguide and the heater. Another potential solution for enhancing the efficiency of T-O phase modulation is the implementation of a deep-etched cavity surrounding the heater and waveguide. This design will allow for the separation of the waveguide and heater from the cladding, effectively isolating them. This approach is applied in [187] and [188] to achieve lower switching power consumption.

2) ELECTRO-OPTIC

Electro-optic (E-O) switching is a method of changing the refractive index of a material using an electric field based on carrier injection control. The E-O controllers are FEOL devices that use p-i-n junction diodes on the sides of ridge waveguide and when the forward bias is applied it triggers the free carrier plasma dispersion effect (FCD) [189] which can change the refractive index of the waveguide. The phase shift can be expressed as a function of carrier density concentration, free-carried absorption, and effective index modulation through Kramers-Kronig relations in Equation 2 [12].

$$\Delta\phi(\lambda, N) = \left(\frac{2\pi}{\lambda} \Delta n(N) + j\Delta\alpha(N)/2 \right) \Gamma L \quad (2)$$

where $\Delta n(N)$ is the effective index change of the waveguide, N is carrier density, and $\Delta\alpha(N)$ is the change of free-carried absorption. The switching time of E-O switches is generally faster than that of T-O switches. E-O switches can switch within nanoseconds to microseconds, depending on the type of material used and the strength of the electric field applied. The power consumption of TO switches is usually in the range of $1\text{--}2\text{ mW}/\pi$. Longer devices enhance power efficiency as they need lower carrier densities. The equation developed by Soref and Bennett [189] that is commonly referenced concerning the change in index caused by free carriers (Equation 3) however, at lower carrier densities the effective index modulation is larger than what this equation predicts [12], [40]. The primary disadvantage of E-O switches is the inherent free carrier absorption (FCA)-induced optical loss. However, their faster speed makes them appealing for various applications [12], [40].

$$\Delta n = -8.8 \times 10^{-22} \Delta N_e - 8.2 \times 10^{-18} (\Delta N_h)^{0.8} \quad (3)$$

Other than silicon, lithium niobate (LiNbO_3) is well known for its large E-O coefficient. LNOI material has the advantage of changing the mode effective index through the E-O effect instead of the plasma dispersion effect of silicon. Besides, the switching time of LNOI switches is usually in the range of femtoseconds [190]. However, the development of LNOI still faces challenges such as costly fabrication techniques [190] and since LN doesn't have a direct band gap similar to silicon integration of a light source wouldn't be feasible [190], [191].

Materials such as Gallium arsenide (GaAs) [192], Indium phosphide (InP) [193], Organic electro-optic materials [194], and Ferroelectric materials [195] are all significant in the field of E-O switching. However, due to the maturity of the silicon fabrication process, silicon stands as a superior option for E-O applications.

B. PHASE-CHANGE MATERIALS

Despite significant advancements in silicon-integrated optical switches, achieving low power consumption, compact design, wide operational range, and minimal insertion loss simultaneously remains challenging. Phase-change materials (PCMs) have emerged as a promising solution in optical switches due to their ability to quickly and reversibly transition between amorphous and crystalline states in response to external stimuli like heat or light. These transitions result in distinct optical properties and self-holding capabilities, which enable control over light transmission and reflection. PCMs can be manipulated through various methods, such as heat, optical, and electrical induction, yielding benefits like fast response times, low power usage, high contrast ratios, and adaptability to smaller dimensions. As a result, PCMs offer significant potential for advancing optical switching applications [14], [196], [197].

Although current photonic integrated circuits primarily rely on inherent T-O and E-O effects for dynamic functions like switching, modulation, and processing, the growing demand for field-programmable photonic circuits is driven

by emerging neuromorphic and quantum computing applications. PCMs such as GST and GSST, initially used in electronic memory, show promise for non-volatile photonic components but can introduce significant losses in the optical domain [14], [198].

Notable PCMs include GeSbTe (GST) [199], [200], AgInSbTe (AIST) [201], $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST2) [202], VO_2 [203], [204], chalcogenides [205], and chalcogenide glasses [206] like As_2S_3 . These materials exhibit various desirable properties, such as good thermal stability, fast switching speeds, and high optical transparency, making them suitable for data storage and emerging photonics and optoelectronics applications [14], [197], [207]. PCM switches are highly compatible with reconfigurable optical networks due to their ability to change states without consuming power during the reconfiguration process, making them an energy-efficient choice for optical network systems [208], [209], [210], [211].

The choice of PCM largely depends on the contrast between the refractive indices of the amorphous and crystalline states. The switching speed between these states is influenced by crystallization speed, which is slower than amorphization. Various studies have explored techniques for improving the switching speed of optical switches, such as incorporating tungsten to decrease resistivity differences or using rapid thermal annealing to introduce phase changes [204], [212], [213], [214].

Waveguide switching, another optical switching method, focuses on/off control of light transmission within the waveguide by incorporating PCM materials within or on top of the silicon waveguide [14], [215]. This approach leverages the unique properties of PCMs, allowing for precise control over light transmission within the waveguide [215]. Fig. 17 demonstrates an example of a waveguide polarization switch described in [216] which is based on an asymmetric Sb_2Se_3 PCM material placed above the Silicon waveguide which can change the polarization state from TE to TM.

In conclusion, PCMs have shown great potential for advancing optical switching applications due to their unique properties, energy efficiency, and adaptability. Continued research and development in this area will likely lead to significant improvements in photonic integrated circuits, paving the way for innovations in neuromorphic and quantum computing applications.

C. ALL OPTICAL SWITCHES

All-optical switching, an emerging technology that enables light to be controlled by light, has become a crucial component in high-speed optical connection networks and integrated logic computing chips. This technology has garnered significant research interest, resulting in advancements across various fields such as nanophotonics [217], integrated optics [217], nonlinear optics [218]. All-optical switching facilitates the modulation of signal-light propagation using a control light, which is achieved through a range of third-order optical nonlinear effects.

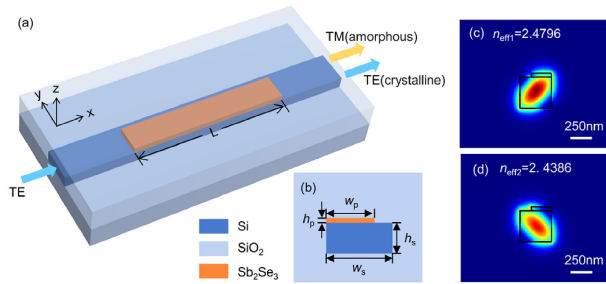


FIGURE 17. (a) A schematic representation of the proposed polarization switch. (b) Cross-sectional view of the polarization rotation area within the polarization switch. (c-d) Depictions of the electric field profiles for the two lowest-order hybridized modes, featuring a-Sb₂Se₃ at a wavelength of 1550 nm [216]. Redrawn from [216]. Licensed under a Creative Commons Attribution (CC BY) License.

All-optical switches can be classified based on performance criteria such as rapid switching time, low threshold control power, exceptional switching efficiency, and compact feature size. The performance of these switches is heavily influenced by the nonlinear optical response properties of the materials used. To create efficient optical switches, it is important to attain both an ultrafast response time and a high third-order nonlinear susceptibility in these materials [219]. Traditional nonlinear media, like conjugated organic materials and semiconductor materials, exhibit limited nonlinear susceptibilities. The challenge lies in balancing the large nonlinear refractive index and ultrafast response time, which are inherently conflicting properties [219], [220].

To address these challenges, researchers are exploring various strategies, such as utilizing new materials [221], and various configurations of metamaterials [222]. This involves enhancing the interaction between light and matter, harnessing slow light effects, and implementing control light transmission in waveguides for practical chip integration applications. Further research is necessary to overcome the remaining challenges and fully unlock the potential of all-optical switching.

Devices employed in all-optical switching are commonly nonlinear photonic microstructures, such as nonlinear photonic crystals and nanocavities, as well as nonlinear plasmonic nanostructures like nonlinear plasmonic nanocavities and nonlinear metamaterials/metasurfaces. All-optical switching depends on the refractive index variation of nonlinear materials, which is induced by the control light through the nonlinear Kerr effect. The intensity of the control light determines the refractive index of the nonlinear materials [219].

The use of photonic crystals in all-optical switching has attracted much attention recently. In [223], authors reported a method of performing ultrafast optical switching utilizing two photonic crystal cavities that were strongly interconnected. By inserting a pump light source into the optical waveguide, they were able to optimize the quality factors and achieve a full oscillation of energy between the two microcavity modes.

Dielectric ring nanocavities have the potential to be used in the construction of all-optical switching devices through the process of side-coupling with a bus signal waveguide. The signal light frequency can be precisely selected at the center of the nanocavity mode. The light signal energy is then coupled into the nanocavity, resulting in the “OFF” state for the all-optical switching as no light signal can transmit through the bus signal waveguide [219].

In [217] authors leveraged the strong and instantaneous quadratic nonlinearity of lithium niobate nanowaveguides for the realization of cavity-free all-optical switching to achieve ultra-low energy consumption down to 80 fJ and a switching time of 46 fs.

Some works use optical heating methods for PCMs to achieve an all-optical switching. As an example in [224], authors proposed an all-optical switch based on PCM. In this work, Ge₂Sb₂Te₅ material is controlled with a 975 nm pulsed laser for tuning a racetrack resonator where they achieved a switch with a response time of 5 μ s.

In [225](Fig.18), authors proposed an all-optical switch based on a graphene-on-silicon nitride MZI switch in which the SiN waveguide is overlayed with a graphene layer that absorbs the pump light to heat the waveguide and causes a phase change in this manner. The size of the proposed device is 0.36 mm² and the extinction ratio of 11 dB.

Jiang et al. [226] recently presented an all-optical MZI switch based on a silicon substrate with a polymer core that leverages the graphene photothermal effect for all-optical switching. In this design, a graphene film is applied to each arm of the MZ interferometer to transform the pump light signal into heat and execute T-O switching on the signal data. Although this switching method eliminates the electronic circuitry needed for traditional E-O and T-O switches, they usually have higher switching time. For example, in [226], the proposed switch exhibits a switching time of 3.7 μ s and a switching power of 6 milliwatts.

In [227], an all-optical tunable microring filter is controlled by using the resonant wavelength shift caused by optical nonlinearities in the silicon microring resonator. This shift changes the frequency difference between the silicon microring resonator’s resonant frequency and the optical carrier’s frequency. More recently, [228] investigated an all-optically tuned hybrid graphene-ultra silicon-rich nitride ring resonator-based add-drop filter for dense wavelength division multiplexing (DWDM) systems. The filter’s tunability was accomplished through an all-optical process using an optical pump to photo-excite the material system’s high nonlinear Kerr effect, applied to DWDM systems in the less-occupied L band. The schematic of this all-optical add-drop filter is shown in Fig. 19.

D. MECHANICAL TUNING

As we discussed before an important category of switches includes MEMS switches. In MEMS switches instead of relying on material properties to create an index shift, the

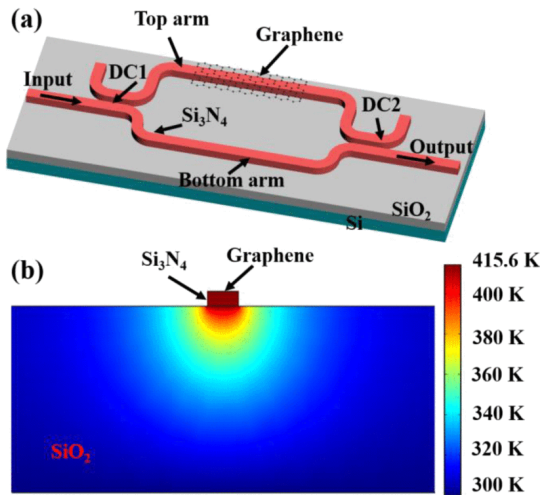


FIGURE 18. a) Schematic of an all-optical MZI switch engine. The switch is composed of a Si_3N_4 waveguide with a graphene layer integrated on top of the waveguide. The device consists of two directional couplers (DC1 and DC2) and a top and bottom arm. Input signals pass through the top and bottom arms, with the graphene enabling dynamic control of the optical signal. (b) Thermal distribution of the device, showing the heat profile in and around the graphene region, with temperatures ranging from 300 K to 415.6 K. The thermal profile is simulated on the SiO_2 substrate with a peak temperature at the graphene- Si_3N_4 interface [225]. Reprinted from [225], Copyright ©2021, IEEE.

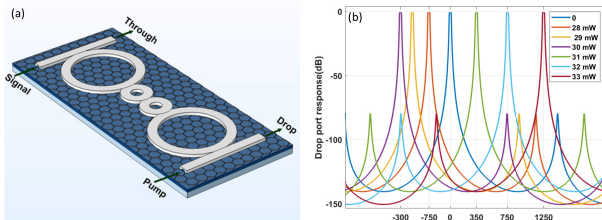


FIGURE 19. (a) 3D schematic of an all-optically tuned add-drop filter. The input signal is injected at the signal port, with resonant wavelengths coupled through the ring resonators and exiting at the drop port, while non-resonant signals exit at the through port. A control pump signal is injected through the add port. (b) Drop port response as a function of the resonance wavelength shift. The original resonance at 1581.5 nm (blue curve) shifts as the injected power varies from 28 to 33 mW. The graphene layer has a thickness of 8 nm and a scattering rate of $\Gamma = 10^{14}/\text{s}$. ©2022 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>) [228].

structures use a mechanical movement to switch the light signal. We can divide the MEMS based on their driven actuation. Internal, electro-static, electromagnetic, piezo-electric, and electro-thermal [229]. Each of these methods has its pros and cons. The passive internal switches are used mainly in sensors for threshold acceleration [229] and are beyond the focus of this paper.

Electro-static actuation uses the attraction between charged objects to cause deformation or displacement. In switches, an applied voltage causes an electrostatic force that moves an electrode, closing or opening the switch [229]. As an example of electro-statically switched waveguides, [230] introduced a 1×5 MEMS silicon nitride photonics switch that uses

electro-static actuators for fine alignment in three switching positions. More recently, [231] reported a dual-slot nano-opto-electro-mechanical (NOEM) phase shifter fabricated in a 90 nm CMOS process that achieves 5.6 radians of phase shift at only 1.08 V bias with < 0.04 dB insertion loss and a 0.26 MHz 3-dB bandwidth, demonstrating that low-voltage, low-loss electro-static tuning is feasible for dense SiPh systems.

Electromagnetic actuation switches function through the motion of a magnetic movable electrode, which is drawn towards a substrate by electromagnetic coils, causing the contacts to meet and activate the circuit. The change in state is managed by applying a bidirectional DC pulse current to the coils. This variety is well-suited for applications that need significant displacements and operate with low voltage to initiate action [229] where they are widely used in radio frequency (RF) applications [229], [232].

In Switch activation occurs via the reverse piezoelectric effect, in which electrical energy applied is transformed into mechanical stress, leading to movement. Commonly utilized materials for this purpose are aluminum nitride (AlN) and lead zirconate titanate (PZT). These switches are characterized by their quick reaction times and consistent performance, capable of handling even minimal shifts [229]. In [233] authors use this effect to introduce a non-volatile memory switch.

Electrothermal switches utilize electrical current to heat and unevenly expand structural materials, creating a bending movement that can activate a switch. Typical designs feature bimorphs and bending beams. Known for their uncomplicated design, they are well-suited for uses where meticulous motion control is less essential [229], [234].

E. CHALLENGES IN OPTICAL TUNING

Optical switching is a crucial technology for modern optical communication systems. There are various methods for achieving optical switchings, such as T-O and electro-optic tuning, all-optical switching, and phase change materials. However, each method has its challenges.

T-O tuning has limitations due to the inherent time it takes for different materials to conduct heat and cause the desired phase shift. This delay can impact the overall latency of the system. Additionally, switches based on T-O tuning consume significant amounts of power, which may lead to increased energy costs and heat management concerns. Localized heating of T-O switches can also cause unwanted interference with adjacent components in the integrated optical environment, leading to performance degradation. Researchers have investigated thermal crosstalk in integrated photonic circuits, demonstrated improved performance using doped heaters, and proposed air-filled trenches to further mitigate crosstalk and enhance phase-tuning efficiency [14], [235].

E-O tuning is limited by the variety of materials that support E-O effects. Furthermore, intrinsic loss imposed on

optical signals due to doping can discourage the use of E-O tuning in certain applications. The voltage required to achieve the desired E-O effect can also pose integration challenges [12], [236].

Although PCMs have shown promising applications in optical tuning, there are still challenges associated with PCM tuning. PCMs often require heating and cooling cycles, which may result in challenges similar to those of T-O switches, such as power consumption and thermal crosstalk. Additionally, some PCMs exhibit degradation in performance after repeated phase transitions, affecting the reliability and longevity of the switch [237]. The phase-transition speed of some PCMs may not be fast enough for certain high-speed applications, which can limit their applicability in ultrafast optical switching. The lack of available fabrication foundries for PCMs also makes their application less practical [14], [238].

All-optical tuning requires materials with large nonlinear susceptibilities and ultrafast response times, which can limit the selection of suitable materials for all-optical switches. Efficient all-optical switching may require high control light power, which can pose challenges in terms of energy consumption and integration with other optical components. All-optical switches can also suffer from crosstalk between signal and control light, as well as noise generated by the nonlinear processes [217], [226].

Different methods of mechanical tuning can face different challenges. Electrostatic switches, while offering low power consumption and fast switching times, can struggle with stiction and electrostatic discharge issues, potentially leading to reliability concerns in sensitive optical environments. Besides their high voltage operation (20-200 V) is not usually feasible through standard CMOS fabrication [229]. Electromagnetic switches are beneficial for their low actuation voltages and large displacement capabilities, but their relatively large size and magnetic interference can be problematic in densely packed optical systems [232]. Piezoelectric switches provide fast response and high precision, yet their reliance on materials like PZT raises concerns over integration compatibility with standard semiconductor processes and potential lead contamination [229], [233]. Electrothermal switches, operating on the principle of thermal expansion, offer simple design and operation; however, their slower response times and potential for thermal crosstalk and drift pose significant challenges for maintaining the strict alignment required in optical switching applications [234]. Overall, while each method has its advantages, the challenges they present necessitate careful consideration and mitigation strategies to ensure effective and reliable optical switching. A comparative summary of these tuning methods is provided in Table 4.

IV. SWITCH FABRIC TOPOLOGY

In the world of networking, topology refers to the way in which devices are connected to form a network. Different topologies have different characteristics, such as scalability,

fault tolerance, and ease of management. Network topologies can be broadly classified into two categories: direct and indirect. Direct topologies, also known as point-to-point topologies, connect nodes directly to each other. While this type of topology offers fast communication, it can be challenging to scale up for larger systems. On the other hand, indirect topologies, also known as switched topologies, use switches or routers to connect nodes, allowing for more efficient use of resources and higher scalability. However, designing an efficient switch is a challenging task that requires careful consideration of various design concerns. In this section, we will review the main design concerns and compare state-of-the-art optical switch topologies in terms of these concerns.

A. OPTICAL NETWORK DESIGN CONCERNS

Switch topology has a crucial impact on the switch performance regarding reliability, footprint, and blocking characteristics [27]. Moreover, technology-related concerns such as waveguide crossing, splitting, and waveguide bending have challenged the design of optical switch topologies. As a result, switch topologies have received much attention in optical switch fabrics. The main concerns in designing topologies for optical switches are summarized in the following.

1) WAVEGUIDE CROSSING

In some switches, it is necessary to use waveguide crossings to achieve a specific routing scheme. However, waveguide crossing affects the optical network negatively, and it is often desirable to avoid waveguide crossings as much as possible. Waveguide crossings can introduce various types of crosstalk and optical losses, which can degrade the performance and power efficiency of the network. These losses and crosstalk not only affect overall network efficiency but also influence device-level parameters such as power consumption and signal integrity. For example, reducing waveguide crossing density can lower insertion losses and improve device speed by minimizing signal distortion. This is because when two waveguides cross, the optical fields of the two waveguides interact, which can result in coupling between the two waveguides. This coupling can cause signal loss, wavelength dependency, and signal distortion, which can degrade the quality of the transmitted signal. Therefore, in optical switch designs, one of the aims is to offer a switch with a minimum number of waveguide crossings. For instance, the N-stage planar [239], shown in Table 5, is designed to remove waveguide crossing and avoid optical losses.

2) BLOCKING/NON-BLOCKING TOPOLOGIES

A non-blocking switch is one in which all inputs can forward data to all outputs simultaneously without any hindrance. Note that a blocking situation is different than a conflict where several inputs request the same output. This means that a non-blocking switch can handle several

TABLE 4. Summary of tuning mechanisms in optical switches.

Tuning Mechanism	Key Principle	Response Time	Power Consumption	Applications and Challenges
T-O	Changes refractive index using heat to alter the optical phase	Microseconds to milliseconds	10–30 mW/ π	Suitable for inter-data center connections and telecom nodes; challenges include high power consumption and thermal crosstalk
Electro-Optic (E-O)	Changes refractive index via electric field using carrier injection (free carrier dispersion effect)	Nanoseconds to microseconds	1–2 mW/ π	Ideal for high-speed networks; challenges include free carrier absorption and voltage integration
Phase-Change Materials (PCMs)	Utilizes reversible transitions between amorphous and crystalline states to control optical properties	Nanoseconds to microseconds	Low (non-volatile)	Promising for field-programmable photonic circuits and neuromorphic applications; challenges include durability and thermal management
All-Optical Switching	Controls light using nonlinear effects (e.g., Kerr effect) without intermediate electronics	Femtoseconds to picoseconds	Depends on control light intensity	Enables ultrafast switching for high-speed optical networks; challenges include high control power and material limitations
Mechanical Tuning (MEMS)	Uses physical movement (e.g., electrostatic, piezoelectric, electromagnetic actuation) to redirect light	Microseconds to milliseconds	Low to moderate	Common in RF and optical sensors; challenges include stiction, high voltage requirements, and integration issues

communication requests simultaneously unless there is a conflict in the requests. Assuming blocking conditions of a specific topology, in a blocking circumstance, waiting should be imposed on some portion of data to solve the blocking. Therefore, a blocking switch topology can negatively impact network traffic and impose latency on packets depending on the network traffic. The latency introduced by blocking conditions is particularly critical for high-speed applications, as it directly impacts the end-to-end response time of devices. This latency can be mitigated by adopting non-blocking topologies like Clos or Benes, which optimize the use of switch cells to balance speed and power consumption.

3) SWITCH FOOTPRINT

The switch footprint comes from the number of switch cells used in the topology. Footprint directly affects not only fabrication cost but also the power consumption of the switch. Minimizing switch footprint is essential for integrating switches into compact photonic chips, where device dimensions are constrained. Smaller footprints not only reduce parasitics, improving overall speed and lowering energy consumption per bit, but also impact tuning mechanisms and device scalability. For example, thermo-optic tuning, which relies on heating localized regions, benefits from reduced dimensions as smaller switch cells require less power to achieve the same phase shift. Similarly, electro-optic switches with compact p-i-n junctions can enhance switching speed and energy efficiency. Devices such as micro-ring resonators and photonic crystal-based switches are particularly advantageous in achieving smaller footprints due to their high integration density and minimal loss. Therefore, it is of great importance to design switches with smaller footprints. For example, Benes topology is designed to offer a non-blocking topology with a minimum number of switch cells and, therefore, has a small footprint and power consumption.

4) SCALABILITY

The number of stages, switch cells, and waveguide crossing in a switch significantly affect performance and power efficiency. Although these drawbacks in switch design can be relatively handled in a switch with a small number of ports, the performance and power efficiency of a switch with a large number of ports can severely suffer as optical losses can be added up. The scalability of a topology is closely tied to the efficiency of its underlying devices. For instance, increasing the port count necessitates more stages in the topology, which can exacerbate thermal issues and increase latency. Advanced materials and device designs, such as low-loss waveguides or optimized micro-ring resonators, can help mitigate these effects. Therefore, a scalable switch is efficient enough in terms of footprint, performance, and power consumption when the number of its ports increases.

5) ROUTING

Depending on the switch topology, different paths might be available to route data from an input to an output. While some switches have a deterministic path from inputs to outputs, high-performance switches offer multiple paths that enable adaptive routing. If the number of available paths between pairs of input to output is large, the performance and energy of the system potentially can be managed efficiently with an appropriate routing algorithm. However, with more path diversity in a switch design, the complexity of routing algorithm and, therefore, its controller increases which can make the design of an efficient controller more challenging [29]. At the device level, this complexity often translates to higher power consumption and slower switching speeds. Implementing adaptive routing algorithms optimized for device parameters, such as the switching speed of micro-ring resonators or the insertion loss of MEMS-based devices, can significantly enhance performance.

B. LATENCY TRADE OFFS

Performance metrics such as latency and bandwidth are significantly impacted by the choice of topology. For instance, Crossbar networks provide low latency due to their direct paths between inputs and outputs, while Fat-Tree topologies offer higher bandwidth through their scalable and fault-tolerant structure. Multistage topologies like Benes balance scalability and low latency, though at the cost of higher power consumption due to additional stages [240]. Topologies like Fat-Tree and Torus are ideal for HPC applications [241], [242] requiring high fault tolerance and scalability. In contrast, Clos networks are commonly used in data centers due to their ability to manage high traffic volumes while ensuring non-blocking communication. Low-latency applications, such as real-time processing, benefit from Crossbar topologies [243], while cost-sensitive systems favor Banyan networks due to their smaller footprint [244]. Latency at the network level is heavily influenced by the switching speed of individual devices. For example, thermo-optic switches, while offering fine control, may introduce higher latency compared to faster electro-optic counterparts. These device-level parameters should be carefully considered when selecting a topology for latency-sensitive applications.

C. SWITCHING NETWORKS TOPOLOGIES

Network topology refers to the arrangement of connections and the flow of information between nodes in a switch communication system. In contemporary systems, switching network topologies are crucial for enabling communication among various components. Optical network topologies exhibit distinct characteristics, such as the number of crossings, switching elements, and switches per path [245], [246].

For example, the crossbar switch has a straightforward structure with direct connections between input and output ports. However, the simplicity of the crossbar topology comes at the cost of requiring a larger number of switching elements, which can increase the overall footprint and power consumption of the network. This trade-off must be balanced against the speed and efficiency requirements of the devices used in the topology. Additionally, it also possesses a high number of crossings, which can lead to increased power loss and crosstalk noise. On the other hand, fat-tree and Clos network switches offer a hierarchical structure with fewer crossings but require more switching elements and have longer paths, potentially causing higher latency and energy consumption [245], [246].

The diverse optical network switch topologies can be classified into five main categories: hierarchical, matrix, multistage, radial/tree-based, and mesh-based. These switch fabric topologies are independent of the underlying multiplexing methods. While WDM (Wavelength Division Multiplexing) is widely used to enhance bandwidth and scalability, the fundamental structure and operation of these topologies

remain applicable to other multiplexing approaches, such as Time-Division Multiplexing (TDM) or Space-Division Multiplexing (SDM). Integrating WDM into these topologies, such as Fat-Tree or Clos, enables simultaneous data transmission over multiple wavelengths, reducing congestion and increasing throughput without altering the core topology. Hierarchical topologies involve multiple layers of switches, matrix topologies feature a matrix structure with direct connections between input and output ports, and multistage topologies incorporate switches arranged across multiple stages. Radial/tree-based topologies utilize tree-like structures, and mesh-based topologies employ mesh-like configurations [246], [247], [248].

Within the hierarchical category, the Fat-Tree topology is notable for its tree-like structure that allows for increased bandwidth as the network grows. This topology is commonly used in data centers and high-performance computing environments due to its scalability and fault tolerance [249]. In contrast, the matrix topology, exemplified by the Crossbar topology, directly connects input and output ports in a grid-like pattern, facilitating simultaneous communication between multiple node pairs. This topology is often found in telephone exchanges and packet-switched networks [87].

Multistage topologies, popular in large-scale networks, include well-known examples such as Banyan [27], Benes [250], Clos [26], and Omega [251] networks. These networks consist of three stages: input, intermediate, and output, and are known for their ability to handle numerous connections with minimal blocking. The input and output stages contain multiple crossbar switches, while the intermediate stage connects these switches in a non-blocking manner. This topology is frequently used in data centers [27], [250], [251].

Meanwhile, radial/tree-based topologies like the Spanning Tree topology connect nodes in a tree-like structure with a central root node, ensuring efficient data transmission and preventing network loops. This topology is often used in Ethernet networks [252]. The Torus topology [253], a mesh-based network example, arranges nodes in a grid and interconnects them in a way that forms a closed loop. This topology is regularly employed in supercomputers and parallel processing systems due to its robustness and low latency.

Banyan switches provide non-blocking communication between inputs and outputs but suffer from non-reconfigurable routing for scalability and reliability purposes. In contrast, the Benes switch, designed with interconnected units, is not only non-blocking but also reconfigurable, improving switch reliability and scalability [254]. Benes topology reduces the number of switch cells while ensuring non-blocking communication [255].

Each category has its own benefits and limitations, with specific switch topologies belonging to these classifications. Designers face challenges in scaling these topologies, such as managing power losses, reducing crosstalk noise, and

minimizing the number of crossings and switch elements, among other concerns [27].

When it comes to switch topology, electronic and optic networks have their own advantages and disadvantages. Therefore, in a hybrid topology, electric and optic inter-connection networks are used together to provide better performance. For example, electronic NoCs are efficient, in comparison with the photonic ones, in cases where small-length packets are sent over short distances. Some papers have put their efforts into balancing the benefits and drawbacks of each technology by incorporating both in hybrid topologies [256], [257].

1) RECENT PROGRESS IN TOPOLOGY OPTIMIZATION (OPTIMIZED SOLUTIONS)

Topology optimization has undergone major developments in recent years, leading to the emergence of novel algorithms, tools, and applications. These advancements have revolutionized the way engineers and designers approach design challenges, enabling them to create optimized solutions that surpass traditional designs in terms of weight reduction, strength enhancement, and overall efficiency.

Researchers have introduced different approaches to overcome some of the problems in the network designs which are borrowed from electronic switch networks. Reducing the number of optical switches in topology can help reduce the optical and electrical power loss this approach was used in [258] to improve the performance of a 4×4 Spanke–Benes network by reducing the number of optical switches in the network where they could achieve 17% less power consumption. Another method of reducing the optical crosstalk and optical loss reduction is by optimization of floor plane design of optical interconnects. In [259] authors discussed an optimization technique to optimize the fat-tree floorplans, aiming to reduce crossings and interconnect length. The number of crossings was reduced by 87% at a cost of longer travel distance.

In [260] authors proposed DRAGON topology a novel high-radix, strictly non-blocking integrated optical switching fabric which has lower loss compared to Crossbar, Benes, and the structure. DRAGON utilizes Micro-Ring Resonators (MRs) as a basic optical switch and it can be seen as two half-crossbars in polar coordinates, consisting of outer and inner loops. The outer loops are located in the outer part of the switching fabric, while the inner loops are in the inner part. Optical signals from any input port can access either outer or inner loops by selecting the appropriate output of the input block and input of the output block.

Another approach discussed in [261] is based on chip partition optimization. The authors present POP, a chip partition optimization platform for silicon-photonic switching fabrics, which effectively reduces power loss, and crosstalk noise, and improves signal integrity, resulting in significant improvements in power loss and signal-to-noise ratio while eliminating transmission failures.

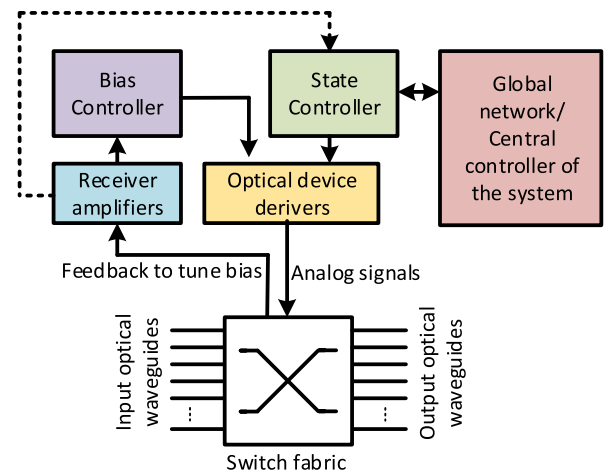


FIGURE 20. Block diagram of the optical switch controller. The input optical waveguides connect to a switch fabric controlled by various system components. The state controller manages the operation of the switch, receiving information from the global network or central controller of the system. Optical device drivers provide analog signals to the switch fabric, while the bias controller adjusts the optical signal levels based on feedback from receiver amplifiers. This feedback loop helps to tune the bias for optimal performance of the optical switch. The system operates with inputs and outputs through optical waveguides.

D. CHALLENGES IN NETWORK TOPOLOGY DESIGN

The design of optical network topologies also faces challenges such as loss, crosstalk, and scalability. Addressing these challenges requires a detailed understanding of the performance parameters of individual devices. For example, reducing crosstalk necessitates precise alignment of waveguides and careful thermal management, while scalability may demand the use of compact, low-power switches to limit power consumption as the network grows. Loss occurs when light signals are absorbed or scattered during transmission, leading to a decrease in signal strength. Crosstalk occurs when the signals from one channel interfere with those from another channel, causing errors in the transmission. Scalability is a significant challenge for optical network topologies, as increasing the number of nodes and links can lead to a more complex structure and higher energy consumption.

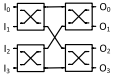
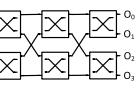
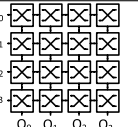
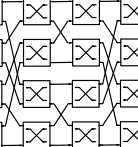
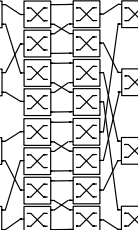
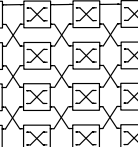
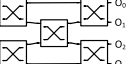
To address these challenges, researchers have proposed various techniques such as optimizing the topology and routing algorithms, using wavelength division multiplexing (WDM) to increase bandwidth, and incorporating advanced materials and devices to reduce loss and crosstalk. Despite these efforts, the design of optical network topologies remains an active research area, and new techniques and solutions continue to be developed to improve the performance and scalability of optical networks.

V. CONTROLLERS

A. INTRODUCTION

To effectively scale optical interconnected networks, it is essential to design modular switch architectures. These

TABLE 5. Switch topology comparison.

	Topology	Advantages	Disadvantages	No. of switches	No. switches in path (No. stages)	Maximum No. Crossing in path
Banyan		Small footprint	blocking switch, low fault tolerance and scalability due to non-reconfigurable architecture	$\frac{N}{2} \log_2 N$	$\log_2 N$	Not found
Benes		Non-blocking, Routing can be reconfigured to improve switch reliability and scalability	Limited scalability due to the need for multiple stages for larger networks	$N \log_2(N) - \frac{N}{2}$	$2 \log_2(N) - 1$	$\frac{2N}{2 \log_2(N) - 2}$
Crossbar		Low loss due to the direct connection between input and output	High cost because of a large number of switch cells	N^2	N	0
Dilated Banyan		Fault-tolerant as it has multiple paths for communication between nodes - high bandwidth as it allows multiple packets to be transmitted simultaneously	Large number of stages and large footprint, High static power consumption to tune this large switch	$2N^2 - 2N$	$2 \log_2(N)$	$\frac{3N}{2 \log_2(N) - 3}$
Dilated Benes		Additional inputs and outputs that allow them to skip over certain stages: Reduce latency and improve path redundancy to enhance fault tolerance	Large number of stages and large footprint	N^2	N	$N - 1$
PILOSS		Uniform loss across all paths	Optical loss and crosstalk due to waveguide crossing	N^2	N	$N - 1$
Planar		Low crosstalk and optical loss due to eliminating waveguide crossing [240]	Optical loss is dependent to the path	$\frac{N}{2}(N - 1)$	$N - 1$	0

modules, which integrate numerous optical switching devices as outlined in the first section of this paper, collectively form a comprehensive switch fabric. In this section, we look into the controllers that play a pivotal role in facilitating communication between the electronic control signals and the analog photonic switch components [12]. By examining these controllers, we gain insights into how electronic instructions are seamlessly translated into the photonic actions that govern the behavior of the optical network.

The controller plays a crucial role in the performance of optical switches since the controller unit takes the main decision over the path. The controller is also employed to handle process and thermal variations of the switch units (e.g., Microring resonator) through a feedback control [262], [263]. The controller is usually designed using an electronic circuit which may include lookup tables (LUTs) to speed up the decision processes. However, optical switches can be

actuated using various methods including optical, electrical, or mechanical means which we discussed in the tuning methods section.

B. CONTROL SCHEME

The controller in an optical network plays a crucial role in the proper functioning, reliability, energy consumption, and performance efficiency of the switch fabric. As shown in Fig 20, it generates analog signals that are used to tune and drive photonic devices of the switch fabric, such as switch cells. The controller's state (configuration) is tuned by the controller state, based on the routing algorithm, communication protocols, and logical interfaces used in other parts of the communication network (Global network in the figure).

To ensure the reliability and performance of the system, the controller of an optical switch is designed to include

two main parts: the bias control unit and the state control unit [12]. The bias control unit receives physical information, which includes some analog signals, from the switch, such as temperature, and adjusts the switch to improve its reliability and performance. In contrast, the state control unit configures the switch based on traffic requests and priorities. Both state and control units use electronic drivers to configure the switch cell properly.

The device drivers are responsible for converting digital signals into corresponding voltage or current signals required to operate the switch fabric, while the receiver amplifiers provide feedback from the switch fabric. These amplifiers may include transimpedance amplifiers, analog-to-digital converters, and low-pass filters [12].

However, as the number of switch cells in an optical switch increases, the latency of path setup, imposed by the state controller, also increases, which can become a bottleneck in the system [110]. Therefore, designers are interested in estimating the latency of path setup to overcome it by employing new architectures and routing algorithms.

In a high-performance controller, the control state unit may also receive feedback from the switch and incorporate the feedback to improve the reliability and energy efficiency of the switch (see the dashed line in Fig 20). For example, in [264] the utilized wavelengths are reconfigured to improve energy efficiency by tuning all MRs according to the utilized wavelengths. Therefore, to set the switch configuration, a well-designed control state unit considers the implemented algorithm and physical information, such as the temperature of switches, conflicts in requests, and the routing algorithm. By taking all these factors into account, the control state unit can make an informed decision about the appropriate configuration for the switch.

C. CONTROLLER OF SOFTWARE DEFINED NETWORKS

In a Software Defined Network (SDN), a centralized software component called the SDN Controller controls the switches [255], [265]. It manages the flow of traffic and configures the switches to forward packets based on the desired network policy. The SDN controller communicates with the switches using the OpenFlow protocol to configure flow tables that help switches make forwarding decisions. The SDN Controller can monitor network performance, detect network faults, and take actions such as rerouting traffic or modifying network policies to address issues. This allows for greater flexibility and programmability in network management. Although the centralized controller in SDN can be more scalable and efficient in the decision-making process to handle traffic dynamically, the controller is relatively a slow method.

D. BIAS CONTROL AND MITIGATION

In high-performance optical networks, one of the major challenges is the precise control of bias in the photonic devices of switches. Various techniques have been proposed

to tackle this challenge. Jayatilleka et al. [266] proposed an automatic bias control of MRs to enable low overhead and fast control of large-scale switches. This approach employs photoconductive heaters in the waveguide to tune the bias MR without any extra contact pads. This method significantly reduces the complexity and cost of the switch fabric.

Moreover, bias control units can also be used to improve the reliability and performance of the switch fabric by compensating for temperature and process variations. In [267] authors proposed an automatic calibration solution to enhance the reliability of photonic pole/zero filters against process and temperature variations. The proposed controller monitors the filter rings' drop ports using photodetectors and calibrates the filters employing microheaters. This approach enables precise tuning of the filters, leading to higher system reliability.

Despite the various solutions proposed for bias control in photonic switch fabrics, the bottleneck of controllers in high-speed switches is controlling the state of the switches rather than the bias. Therefore, researchers are interested in developing controllers that can efficiently control the state of switches in high-speed photonic networks.

Overall, the development of efficient and reliable controllers for bias control in photonic switch fabrics is crucial for the advancement of high-speed communication networks. Future research in this area should focus on developing controllers that can effectively control the state of switches and leverage the benefits of machine learning techniques and hybrid control systems.

E. CONTROLLER DESIGN CONCERNS

Optical switches have emerged as a revolutionary technology that plays a pivotal role in modern high-speed communication networks. By leveraging the enormous capacity and speed of optical signals, these switches have unlocked new possibilities for meeting the ever-increasing demands for high-performance data transmission. However, the design and implementation of effective controllers for optical switches come with a myriad of challenges. In this introduction, we will outline some of the most critical controller design concerns, including latency, integration, dynamic traffic demands, heterogeneous network elements, scalability, fault tolerance, and security [12], [28], [268].

1) LATENCY

The performance of optical switch fabrics is often bottlenecked by the latency of the controllers. Despite the device-level optical switches boasting response times in the scale of nanoseconds to microseconds, the controllers typically exhibit higher response times, which can significantly degrade the performance of the switch fabric. This disparity highlights the need for improvements in the controller design [28].

One potential avenue for enhancing optical interconnection networks lies in minimizing the latency of the

optical switching controller, which would have a considerable impact on the network's overall performance. Rapid decision-making for switch configurations, coupled with efficient conflict resolution, is crucial to achieving higher overall performance. By addressing these latency-related concerns, optical networks can capitalize on the inherent speed advantages of optical switches and provide more seamless communication experiences [12], [28].

2) INTEGRATION

Off-chip controllers typically have more processing power and memory than on-chip controllers, which can allow them to perform more complex operations and handle larger amounts of data. They are also often more flexible and easier to upgrade, as they can be replaced without having to modify the switch itself. However, they may require additional power and space to operate, which can be a disadvantage in certain applications.

On-chip controllers, on the other hand, are more compact and energy-efficient than off-chip controllers, as they are integrated into the same chip as the optical switch. This can result in lower power consumption, lower cost, and a smaller overall footprint. However, on-chip controllers may have limited processing power and memory, which can restrict their ability to perform complex operations or handle large amounts of data. In addition, upgrading an on-chip controller may require modifying the entire chip, which can be more challenging than upgrading an off-chip controller.

3) DYNAMIC TRAFFIC DEMANDS

The traffic demands in optical networks can change rapidly due to various reasons, such as bursty data, sudden changes in user behavior, etc. This can result in congestion and packet loss, and therefore, the controller should be able to adapt to the changing traffic conditions. The traffic patterns in optical networks can be highly dynamic, with traffic demands varying rapidly and unpredictably. This makes it challenging to design a controller that can quickly adapt to changing network conditions and optimize traffic flow accordingly.

4) HETEROGENEOUS NETWORK ELEMENTS

Optical networks are composed of a diverse array of network elements, such as optical switches, routers, and amplifiers, each with different characteristics like switching speeds, capacities, and power requirements. Managing these heterogeneous components effectively is crucial for achieving the desired network performance and ensuring seamless communication experiences.

The controller should be designed to efficiently coordinate varying switching speeds, ensuring smooth traffic flow and minimizing congestion. This may involve intelligent traffic prioritization, dynamic load balancing, and real-time adjustments in response to changing network conditions. Furthermore, the controller must be capable of managing disparate capacities by effectively allocating resources and routing traffic to avoid bottlenecks and overloads. This

can be achieved through advanced routing algorithms and capacity-aware traffic engineering techniques.

In terms of energy efficiency and resource management, the controller should optimize the usage of network elements with different power requirements to minimize energy consumption while maintaining the desired network performance. This may include incorporating power-aware algorithms, dynamic power management strategies, and leveraging energy-efficient network elements when possible. Ensuring compatibility and interoperability between diverse components is also essential, involving support for multiple protocols, interfaces, and communication standards, as well as seamless integration between legacy and next-generation network elements [269], [270].

5) SCALABILITY

Optical networks often experience rapid growth in size and complexity due to the ever-increasing demands for high-speed communication and data transmission. As a result, designing a controller that can effectively scale up and manage the increasing number of network elements and traffic demands without compromising network performance is crucial. This task poses unique challenges, especially when optimizing traffic flow in complex systems with numerous interconnected nodes [271], [272].

To address these challenges, several strategies can be employed. One such approach is the distributed control architecture [28], which divides the control plane into multiple interconnected controllers, each managing a subset of network elements. This division helps alleviate controller load and ensures better resource management. Additionally, to handle increased traffic demands, the controller must dynamically balance loads and allocate resources efficiently using advanced routing algorithms, traffic engineering techniques, and real-time network monitoring [271].

Network partitioning and virtualization can further enhance scalability by dividing the network into smaller segments or virtual domains, allowing for more granular control without impacting overall performance. Incorporating adaptive algorithms and machine learning techniques enables the controller to learn and adapt to changing network conditions, leading to improved resource allocation and network performance. Finally, establishing a comprehensive monitoring system and defining key performance metrics facilitate more informed decision-making and resource allocation by identifying bottlenecks and areas for improvement [273].

6) FAULT-TOLERANCE

Optical switches are susceptible to various sources of faults, such as process variation and thermal variation [274]. The controller should be able to detect and recover from these faults quickly to ensure the network's uninterrupted operation. Effective fault-tolerance requires the controller to quickly identify anomalies, diagnose their causes, and

implement corrective actions, such as reconfiguring optical paths, compensating for thermal drift, or bypassing faulty components. By integrating redundancy strategies, predictive monitoring, and real-time feedback loops, controllers can proactively address potential failures, ensuring seamless operation and minimal disruption to network performance.

Fault modeling and mitigation in optical switches involve addressing various types of failures, including channel faults, link faults, switch faults, and node faults [275]. For example, channel faults caused by wavelength degradation or optical signal impairments may require controllers to dynamically adjust wavelength allocations or reroute traffic using available spare channels. Similarly, node faults can necessitate source-based restoration strategies where traffic is rerouted globally, while link-based restoration can handle localized faults through immediate reconfiguration [275].

Fault modeling and mitigation remain challenging due to the dynamic and diverse nature of faults in optical systems.

7) SECURITY

Optical switches can be vulnerable to attacks, such as eavesdropping, interception, and denial-of-service (DoS). The controller should be able to implement security measures to protect the network from these threats. Optical networks are vulnerable to a range of security threats, including eavesdropping, interception, and tampering. Designing a controller that can effectively detect and respond to these threats is essential for ensuring the security and reliability of the network.

8) HIGH-SPEED DATA RATES

Optical switches operate at high data rates, which can make it difficult to design a controller that can handle the high volume of traffic and process data in real-time. Therefore, controller and/or the connection between the controller and switch, can be a bottleneck. For example, if the switch fabric includes many switch cells, the controller is supposed to control all the cells simultaneously, and the network/connection between the controller and the switch should handle a high data rates with low latency. This necessitates an intra-chip controlling mechanism for a high-speed switch where frequent reconfiguration is required. For instance, in the 2.5D architectures proposed in [211], [264], [276], [277], and [278], the controller is placed on the chiplets to control optical devices on the interposer using micro bump connections, which provide relatively a high bandwidth and low latency link between the chiplets and the interposer.

F. CENTRALIZED, DISTRIBUTED, MULTI-AGENT CONTROLLERS

Centralized management in which the whole switch is configured by one controller unit is faster in comparison with the distributed one. In [279], a centralized controller is designed for a MZI-based switch. To speed up the routing decision mechanism and resolve configuration conflicts in one cycle,

look-up tables (LUTs) are employed. Göhring et al. [280] proposed a centralized controller for optical integrated networks to decrease complexity and improve latency. The proposed controller has a low complexity due to its centralized design and a global view of switches. In order to improve the latency of controller decisions, a connectivity matrix is employed to solve conflict situations in which there are several requests for the same resource. The round-robin algorithm is implemented using a first-in-first-out (FIFO) buffer to manage access to resources in case of any conflict. This controller offers low latency management of the optical interconnection network mainly due to its lookup table. However, employing a lookup table makes the design not scalable, and, therefore, the application of such a controller is limited to be utilized in a small interconnection network. In [281], the authors proposed a lookup table-based controller in which the routing for each path is done offline, and using some optimization (acceleration techniques) their method makes the table more efficient. The latency of the controller in the worst case is smaller than 5 ns which is quite small. However, the controller is not scalable because of the implemented lookup table. In general, a centralized controller is not scalable to the size of the network.

In [282], a centralized controller is designed on a separate chip to control both the bias and state of the switch. However, the controller is relatively slow which limits its application to the low frequency of switching.

Yang et al. [283] proposed a distributed controller that is scalable for their specified large design. However, the proposed controller is topology-constrained and can be used in their interconnection network which is proposed in their paper. Li et al. in [284] proposed a distributed controller for an H-tree interconnection network which is scalable to be used for a large H-tree interconnection network. However, for some decisions, the controller takes 3 cycles to configure the path which is not efficient.

G. AI-BASED CONTROLLERS

Traditionally, fixed algorithms or pre-determined routing tables have been used to design controllers for optical switches. However, as optical switches become more complex to support larger-scale systems with high traffic and bandwidth demands, traditional controllers are no longer efficient. To address this issue, AI-based controllers have been developed that can intelligently evaluate switch circumstances, including power loss, traffic congestion, and switch utilization, to configure the switch for higher performance [273].

In [285], artificial neural networks (ANNs) are employed to predict the blocking probability in optical networks. The paper uses the topological property of optical networks to represent the network's capacity and distribute traffic load accordingly. This approach has shown that ANNs can significantly improve the accuracy of blocking prediction in optical networks while reducing the time required for

prediction calculations. However, the proposed method may not be suitable for all real-time applications since the learning engine employs an offline discrete-event simulator for the training process.

Another approach for optimizing optical networks is Ant Colony Optimization (ACO), as demonstrated in [286]. ACO is a meta-heuristic optimization algorithm that has found broad application in the routing of interconnection networks. ACO-based routing algorithms have been shown to be useful not only for energy minimization but also for handling network congestion and improving latency. By using ACO to find paths with minimized energy consumption, the paper demonstrates that it is possible to improve the efficiency of optical networks.

Recent work has moved beyond shallow ANNs toward deep-learning-driven control. Fully-connected neural networks, graph neural networks, and hybrid architectures now support tasks such as traffic prediction, alarm correlation, and anomaly detection, enabling real-time responsiveness and fine-grained telemetry integration [287], [288]. Composable machine-learning designs further improve generalization across unseen topologies by separating loading, recursive feature extraction, and read-out stages [289].

Deep reinforcement learning (DRL) has emerged as a powerful tool for dynamic resource allocation. Shabka et al. demonstrated a DRL-based auto-tuning of PID loops in MEMS circuit switches, reducing re-configuration latency by 22 % [290]. At the network layer, DRL agents have been shown to restore lightpaths after link failures [291], to provision routes, modulation formats, and spectra in elastic optical networks with lower blocking than heuristic baselines [292], and—when aided by a diffusion-based generative model—to achieve the state of the art in routing and wavelength assignment [293].

Hierarchical SDN/AI orchestration is also gaining traction. Castoldi et al. propose integrating streaming telemetry with intent-based SDN controllers, where centralized AI handles network-level optimization and on-board AI manages equipment-level tuning [294]. Such layering balances computational load while preserving scalability.

Despite these advances, recent benchmarking reveals that well-tuned heuristics can still match or surpass many published DRL methods, underscoring the need for reproducible baselines and transparent evaluation [295]. Moreover, non-technological barriers—data-sharing restrictions, explainability concerns, and lack of standardization—remain critical hurdles to deployment [296]. Open-source platforms such as the FUSION simulator aim to accelerate progress by providing a common test-bed with native ML/RL support [297]. Future developments may include federated learning to minimize data transfer overheads, energy-aware RL that considers physical tuning costs, and hybrid agents capable of joint device- and network-level optimization.

Beyond the C-band, Ben Terki et al. extend DRL to multi-band optical networks and achieve up to an 80 %

reduction in blocking probability compared with state-of-the-art heuristics [298]. Distributed learning is being tackled through a federated-learning SDON architecture that keeps model training local to each node, lowering bandwidth and privacy overheads while still realising global convergence [299]. Looking ahead, a modular AI control framework spanning optical and wireless domains has been proposed for 6G, enabling unified, zero-touch management across heterogeneous layers [300].

Collectively, these efforts indicate a transition from heuristic, manually engineered control toward autonomous, data-driven optical switching systems capable of adapting to ever-growing performance and reliability demands.

VI. CONCLUSION

In conclusion, this comprehensive survey paper provides a holistic overview of optical switches, highlighting their critical role in addressing the increasing demands of modern applications in data transmission and processing. Traditional electrical switches are becoming insufficient for supporting the high data rates, low latency, and scalability requirements of applications such as big data analytics, data center networks, high-performance computing (HPC), artificial intelligence (AI), 5G, Internet of Things (IoT), and cloud computing. Optical switches, with their high bandwidth, energy efficiency, and fast reconfigurability, present a transformative solution to these challenges.

This paper explores a wide array of topics, ranging from the types of optical switching devices and their underlying technologies to network topologies and control mechanisms. We analyzed the strengths and limitations of various devices, such as silicon-photonics, III-V photonic, MEMS, and LCOS switches, and examined tuning mechanisms like electro-optic, thermo-optic, and phase-change materials. The discussion extends to direct and indirect network topologies, offering insights into their design, optimization, and challenges in achieving scalability, low loss, and low crosstalk.

Furthermore, the survey looks into the role of optical switches in enabling emerging applications. AI workloads, for instance, rely on optical switches to handle the massive data exchange required for training large-scale models, as well as to achieve low-latency inference in distributed systems. IoT ecosystems benefit from the seamless connectivity provided by optical switches, which support dynamic traffic management for real-time edge-to-cloud communication. Similarly, big data applications require the high-throughput and low-latency capabilities of optical switches to process and transfer petabytes of information efficiently across distributed systems.

We also examined control schemes, including centralized, distributed, and multi-agent solutions, along with the integration of advanced machine learning techniques into controller design and routing systems. Recent advancements, such as reinforcement learning for dynamic resource allocation and

hybrid AI models that balance centralized and distributed processing, demonstrate the growing intersection of optical switching technologies with AI-driven innovations. These developments address challenges in scalability, fault tolerance, and real-time performance, which are critical for large-scale networks.

This survey aims to serve as a valuable resource for researchers and professionals in the field, offering a comprehensive understanding of the current state of the art in optical switching technologies. By identifying less explored areas, such as the integration of federated learning in distributed optical networks or the use of hybrid AI architectures for fault-tolerant systems, we aim to inspire further research and innovation. As applications in AI, IoT, and big data continue to grow, the development of advanced optical switches will play an essential role in shaping the future of data-driven technologies.

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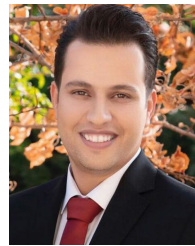
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